

# RRB + SSC MATHS

**Sirf 53 Formulas**

Sawaal padho

→

PEHCHAN shabd dhoondo

→

Value rakho

→

Answer

3 books ke 13,497 sawaal: RRB English · RRB Hindi · SSC Chapterwise

Har formula ka example computer se check kiya ✓

## Kaise Use Karein

**1**

**Sawaal padho**

Usme koi khaas shabd dhoondo  
(jaise “increased by 20%”)



**2**

**PEHCHAN**

Agle page ki list mein shabd  
dekho → formula mil gaya  
(PC-2)



**3**

**Value rakho**

Card ki table mein multi-julti  
line chuno, numbers daalo →  
**answer**

**Example:** “Ek cheez ₹500 ki thi, 20% mehngi hui. Naya daam?”

Shabd: **mehngi** / **badha** → **PC-2** →  $\text{Naya} = 500 \times 120/100 = \text{₹600}$

## Koi formula yaad na aaye to ye 5 trick

1. **100 maan lo** — %, profit, discount, income
2. **LCM = kul kaam** — kaam, nal-tanki
3. **1 hissa nikaalo** — ratio, umar, partnership
4. **Value daalo** — algebra:  $x = 1, 2$  · trigo:  $\theta = 45^\circ, 30^\circ$
5. **Options daalo** — har option ko sawaal mein rakh ke check

## PEHCHAN LIST — kaunsa formula?

Sawaal mein ye shabd dikhe → wo formula lagao (page number saath mein)

| Formula                                                         | Sawaal mein ye shabd                                                                                                        | Page      |
|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>1. Number System</b> संख्या पद्धति                           |                                                                                                                             |           |
| <b>NS-1</b>                                                     | divisible / विभाज्य · remainder / शेषफल · unit digit / इकाई अंक · kitna jodein / ghatayein · sabse bada / chhota n-digit    | <b>5</b>  |
| <b>NS-2</b>                                                     | how many numbers / कितनी संख्याएँ · factors / divisors / गुणखंड · prime / co-prime · zeros at the end (n!) · sum of factors | <b>5</b>  |
| <b>NS-3</b>                                                     | sum and difference / योग और अंतर · consecutive numbers / क्रमागत · digits reversed · $1 + 2 + \dots + n$ · sum of squares   | <b>6</b>  |
| <b>2. Decimal Fractions</b> दशमलव भिन्न                         |                                                                                                                             |           |
| <b>DF-1</b>                                                     | p/q form · bar (recurring) decimal · terminating · reciprocal / व्युत्क्रम · lowest form                                    | <b>7</b>  |
| <b>DF-2</b>                                                     | largest / smallest fraction · ascending / descending · kya jodein · bacha hissa · numerator / denominator                   | <b>7</b>  |
| <b>3. Indices and Surds</b> घातांक तथा करणी                     |                                                                                                                             |           |
| <b>IS-1</b>                                                     | power / exponent / घात · find x in $a^x$ · $\sqrt{\quad}$ inside $\sqrt{\quad}$ · $\sqrt{x}$ equation                       | <b>8</b>  |
| <b>IS-2</b>                                                     | square root / वर्गमूल · $\sqrt{\quad}$ of decimal · surd / करणी · rationalise · which is greater ( $\sqrt{\quad}$ )         | <b>8</b>  |
| <b>4. Simplification</b> सरलीकरण                                |                                                                                                                             |           |
| <b>SIM-1</b>                                                    | simplify / सरल कीजिए · find ? · approximate value · brackets, of                                                            | <b>9</b>  |
| <b>SIM-2</b>                                                    | same numbers repeat · $a^2 - b^2$ · $x : y$ diya · $\sqrt{(n + \sqrt{(n + \dots))}$                                         | <b>9</b>  |
| <b>5. LCM &amp; HCF</b> लघुत्तम समापवर्त्य एवं महत्तम समापवर्तक |                                                                                                                             |           |
| <b>LH-1</b>                                                     | HCF / म.स. · LCM / ल.स. · product of two numbers · ratio and HCF · HCF of fractions                                         | <b>10</b> |
| <b>LH-2</b>                                                     | least number leaving remainder · greatest number which divides · bells ring together / घंटियाँ · largest tile / container   | <b>10</b> |

|                                                        |                                                                                                                             |           |
|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>6. Percentage</b> प्रतिशतता                         |                                                                                                                             |           |
| <b>PC-1</b>                                            | x% of · what percent / कितना प्रतिशत · pass marks / fail by · income, expenditure, savings · votes / election               | <b>11</b> |
| <b>PC-2</b>                                            | increase / decrease · बढ़ा / घटा · successive / do baar · population / जनसंख्या · consumption / खपत                         | <b>11</b> |
| <b>7. Profit &amp; Loss</b> लाभ और हानि                |                                                                                                                             |           |
| <b>PL-1</b>                                            | cost price / क्रय मूल्य · selling price / विक्रय मूल्य · profit % / loss % · marked price + discount · sold for ₹. . . more | <b>12</b> |
| <b>PL-2</b>                                            | SP of x articles = CP of y · a for ₹x, b for ₹y · false weight / गलत बाट · part sold at x%, rest at y%                      | <b>12</b> |
| <b>8. Discount</b> बट्टा / छूट                         |                                                                                                                             |           |
| <b>DS-1</b>                                            | marked price / अंकित मूल्य · discount / छूट · successive discount · buy x get y free                                        | <b>14</b> |
| <b>9. Ratio &amp; Proportion</b> अनुपात और समानुपात    |                                                                                                                             |           |
| <b>RP-1</b>                                            | divide in ratio / अनुपात में बाँटो · A : B and B : C · x% of A = y% of B · coins / सिक्के · income ratio, expenditure ratio | <b>15</b> |
| <b>RP-2</b>                                            | proportion / समानुपात · mean / third / fourth proportional · number added to make ratio · (x + y) : (x - y)                 | <b>15</b> |
| <b>10. Partnership</b> साझेदारी                        |                                                                                                                             |           |
| <b>PT-1</b>                                            | partnership / साझेदारी · invested / निवेश · share of profit · joined after . . . months                                     | <b>16</b> |
| <b>11. Work &amp; Time</b> समय और कार्य                |                                                                                                                             |           |
| <b>WT-1</b>                                            | work / काम · together / साथ मिलकर · alone / अकेले · efficient · left after / beech mein chhoda · wages / मजदूरी             | <b>17</b> |
| <b>WT-2</b>                                            | men, women / पुरुष, महिलाएँ · days, hours per day · food lasts / भोजन                                                       | <b>17</b> |
| <b>12. Alligation / Mixture</b> मिश्रण / एलीगेशन       |                                                                                                                             |           |
| <b>MX-1</b>                                            | in what ratio mixed / किस अनुपात में · mixture price · mean price                                                           | <b>18</b> |
| <b>MX-2</b>                                            | water added / पानी मिलाया · evaporated · taken out and replaced                                                             | <b>18</b> |
| <b>13. Pipe &amp; Cistern</b> नल और टंकी               |                                                                                                                             |           |
| <b>PP-1</b>                                            | pipe, tap / नल · tank / टंकी · fill + empty · leak / रिसाव · opened alternately                                             | <b>19</b> |
| <b>14. Simple Interest</b> साधारण ब्याज                |                                                                                                                             |           |
| <b>SI-1</b>                                            | simple interest / साधारण ब्याज · amount / मिश्रधन · rate, time · becomes double / n times                                   | <b>20</b> |
| <b>15. Compound Interest</b> चक्रवृद्धि ब्याज          |                                                                                                                             |           |
| <b>CI-1</b>                                            | compound interest / चक्रवृद्धि · compounded half-yearly / quarterly · amount after n years · instalments                    | <b>21</b> |
| <b>CI-2</b>                                            | difference between CI and SI · CI and SI for 2 years                                                                        | <b>21</b> |
| <b>16. Problems on Age</b> आयु संबंधी प्रश्न           |                                                                                                                             |           |
| <b>AG-1</b>                                            | age / आयु · years ago / वर्ष पहले · after . . . years / वर्ष बाद · ratio of ages                                            | <b>22</b> |
| <b>17. Average</b> औसत                                 |                                                                                                                             |           |
| <b>AV-1</b>                                            | average / औसत · joins / leaves / replaced · runs / marks · consecutive numbers                                              | <b>23</b> |
| <b>18. Speed, Time &amp; Distance</b> चाल, समय और दूरी |                                                                                                                             |           |
| <b>ST-1</b>                                            | speed, distance, time / चाल, दूरी, समय · km/h, m/s · late / early / देर, जल्दी · usual speed ka 3/4                         | <b>24</b> |
| <b>ST-2</b>                                            | average speed / औसत चाल · towards each other / आमने-सामने · same direction / पीछा · meet / overtake                         | <b>24</b> |
| <b>19. Train</b> रेलगाड़ी                              |                                                                                                                             |           |
| <b>TR-1</b>                                            | train / रेलगाड़ी · pole, man, tree / खंभा · platform, bridge, tunnel · two trains / दो ट्रेनें                              | <b>25</b> |
| <b>20. Boat &amp; Stream</b> नाव और धारा               |                                                                                                                             |           |
| <b>BS-1</b>                                            | boat, stream / नाव, धारा · downstream / upstream · still water / शांत जल                                                    | <b>26</b> |

|                                                                |                                                                                                                                                           |           |
|----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>21. Mensuration</b> क्षेत्रमिति                             |                                                                                                                                                           |           |
| <b>MN-1</b>                                                    | area / क्षेत्रफल · perimeter / परिमाप · triangle, rectangle, square, rhombus · circle, wheel, path · hexagon                                              | <b>27</b> |
| <b>MN-2</b>                                                    | volume / आयतन · surface area / पृष्ठ · cube, cuboid, cylinder · cone, sphere, hemisphere · melted and recast / पिघलाकर · prism, pyramid                   | <b>28</b> |
| <b>MN-3</b>                                                    | radius doubled · side increased by x% · ratio of volumes / areas                                                                                          | <b>28</b> |
| <b>22. Algebra</b> बीजगणित                                     |                                                                                                                                                           |           |
| <b>AL-1</b>                                                    | $x + 1/x = \dots$ · $a + b = \dots$ , $ab = \dots$ · $a + b + c = 0$ · $x^3 + y^3 + z^3 - 3xyz$ · simplify / factorise                                    | <b>29</b> |
| <b>AL-2</b>                                                    | solve, find x and y · roots / मूल · quadratic · remainder when divided by $(x - a)$ · factor                                                              | <b>29</b> |
| <b>AL-3</b>                                                    | nth term · sum of n terms · progression / श्रेणी                                                                                                          | <b>30</b> |
| <b>AL-4</b>                                                    | both / दोनों · neither / कोई नहीं · only / केवल                                                                                                           | <b>30</b> |
| <b>23. Trigonometry</b> त्रिकोणमिति                            |                                                                                                                                                           |           |
| <b>TG-1</b>                                                    | sin, cos, tan ... ka maan · simplify / prove / equals · $\sec \theta + \tan \theta = k$ · if $\tan \theta = p/q$ · maximum / minimum value                | <b>31</b> |
| <b>TG-2</b>                                                    | $\sin 37^\circ / \cos 53^\circ$ · $\tan 1^\circ \tan 2^\circ \dots \tan 89^\circ$ · $\sin A = \cos B$ · $\sin(A + B)$ , $\cos 2\theta$ · radian           | <b>31</b> |
| <b>TG-3</b>                                                    | angle of elevation / उन्नयन कोण · angle of depression / अवनमन · tower, shadow, ladder, kite                                                               | <b>32</b> |
| <b>24. Co-ordinate Geometry</b> निर्देशांक ज्यामिति            |                                                                                                                                                           |           |
| <b>CG-1</b>                                                    | distance between points / दूरी · midpoint / ratio · area of triangle (x, y) · collinear                                                                   | <b>33</b> |
| <b>CG-2</b>                                                    | slope / ढाल · equation of line · parallel / perpendicular · intercept, area with axes · circle centre, radius                                             | <b>33</b> |
| <b>25. Geometry</b> ज्यामिति                                   |                                                                                                                                                           |           |
| <b>GE-1</b>                                                    | angle / कोण · parallel lines · triangle angles, exterior angle · bisectors meet at O · polygon: interior / exterior · parallelogram, cyclic quadrilateral | <b>34</b> |
| <b>GE-2</b>                                                    | Pythagoras · similar triangles / समरूप · angle bisector · median, centroid / माधिका · perpendicular on hypotenuse · inradius, circumradius                | <b>34</b> |
| <b>GE-3</b>                                                    | chord / जीवा · tangent / स्पर्श रेखा · angle at centre · two circles, common tangent / chord · secant meets outside                                       | <b>35</b> |
| <b>26. Statistics &amp; Probability</b> सांख्यिकी और प्रायिकता |                                                                                                                                                           |           |
| <b>SP-1</b>                                                    | mean / माध्य · median / माधिका · mode / बहुलक · frequency table                                                                                           | <b>36</b> |
| <b>SP-2</b>                                                    | range / परास · variance / प्रसरण · standard deviation / मानक विचलन                                                                                        | <b>36</b> |
| <b>SP-3</b>                                                    | probability / प्रायिकता · coin, dice, cards · balls drawn                                                                                                 | <b>36</b> |
| <b>27. Data Interpretation</b> आंकड़ों का विश्लेषण             |                                                                                                                                                           |           |
| <b>DI-1</b>                                                    | pie chart · bar graph · table / तालिका · line graph · percentage increase, ratio, average                                                                 | <b>38</b> |
| <b>28. Miscellaneous</b> विविध                                 |                                                                                                                                                           |           |
| <b>MS-1</b>                                                    | of the remaining / बाकी का · map scale 1 : n · unitary / ऐकिक · puzzle: doubling, digits                                                                  | <b>39</b> |

# 1 Number System संख्या पद्धति

## NS-1 BHAAG – SHESH

**PEHCHAN** divisible / विभाज्य · remainder / शेषफल · unit digit / इकाई अंक · kitna jodein / ghatayein · sabse bada / chhota n-digit

$$N = D \times Q + R$$

N = number · D = jisse bhaag · Q = bhaagfal · R = shesh

| Sawaal mein ye ho                     | Formula / kya karo                                                                                                              | Example: value rakho → answer                                          |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| ① Divisible hai? / digit x            | 2, 5 → last digit · 3, 9 → digits ka jod · 4 / 8 → last 2 / 3 digit · 11 → (odd – even jagah) = 0 ya $11 \cdot 72 = 8 \times 9$ | $8712 \div 11 \rightarrow (8+1) - (7+2) = 0 \checkmark$                |
| ② Bade guna / power ka shesh          | Har number ki jagah uska shesh rakho, phir guna                                                                                 | $(17 \times 23) \div 5 \rightarrow 2 \times 3 = 6 \rightarrow$ shesh 1 |
| ③ Kitna jodein / ghatayein            | Jodo = D – R · Ghatao = R                                                                                                       | $100 \div 7, R = 2 \rightarrow$ jodo 5, ghatao 2                       |
| ④ Sabse bada / chhota n-digit jo kate | Bada = $99 \dots 9 - R$ · Chhota = $10 \dots 0 + (D - R)$                                                                       | 3-digit, ÷ 7: $999 - 5 = 994$                                          |
| ⑤ Unit digit of $a^n$                 | $n \div 4 \rightarrow$ bacha r (0 ho to 4) → (last digit) <sup>r</sup>                                                          | $7^{23}$ : $23 \div 4$ bacha 3 → $7^3 = 343 \rightarrow 3$             |
| ⑥ $a^n - b^n$ kis se kate             | $a^n - b^n \rightarrow (a - b)$ se hamesha, n even ho to $(a + b)$ se bhi · $a^n + b^n$ , n odd → $(a + b)$ se                  | $17^{20} - 7^{20} \rightarrow 10$ aur 24 dono se                       |

BOOK SE (asli sawaal)

- ① **SSC · Number System Q9** 8A5146B, 88 se divisible ⇒  $A \times B = ?$   
 $8 \rightarrow$  last 3 digit 46B ⇒  $B = 4 \cdot 11 \rightarrow (8+5+4+4) - (A+1+6) = 0/11 \Rightarrow A = 3 \Rightarrow AB = 12 \checkmark$  (b)
- ① **RRB English · Ch 1 Q1** 9 aur 11 dono se divisible?  
 10098: digit-sum 18 ✓(9);  $(8 + 0 + 1) - (9 + 0) = 0 \checkmark$ (11) ✓ (b)

## NS-2 GINTI + FACTORS

**PEHCHAN** how many numbers / कितनी संख्याएँ · factors / divisors / गुणनखंड · prime / co-prime · zeros at the end (n!) · sum of factors

$$N = p^a q^b \Rightarrow \text{Factors} = (a + 1)(b + 1)$$

p, q = prime · a, b = unki power

| Sawaal mein ye ho               | Formula / kya karo                                                                                                        | Example: value rakho → answer                    |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| ① A se B tak kitne d se katenge | $\lfloor B/d \rfloor - \lfloor (A-1)/d \rfloor$                                                                           | 1–100 mein 7 se: $100 \div 7 = 14$               |
| ② Kitne factors                 | Powers + 1 karke guna · perfect square factors: $(\lfloor a/2 \rfloor + 1) \dots$ cube: $(\lfloor a/3 \rfloor + 1) \dots$ | $72 = 2^3 \cdot 3^2 \rightarrow 4 \times 3 = 12$ |
| ③ Factors ka jod                | $(1 + p + \dots + p^a)(1 + q + \dots + q^b)$                                                                              | $12 = 2^2 \cdot 3 \rightarrow (1+2+4)(1+3) = 28$ |
| ④ n! ke end mein kitne zero     | $\lfloor n/5 \rfloor + \lfloor n/25 \rfloor + \dots$                                                                      | $100! \rightarrow 20 + 4 = 24$                   |
| ⑤ Prime hai? / co-prime         | $\sqrt{N}$ tak ke prime se bhaag · co-prime ⇔ HCF = 1                                                                     | 97: 2, 3, 5, 7 se nahi kata → prime              |

**BOOK SE (asli sawaal)**

⑤ **SSC • Number System Q152** 10 se 100 ke beech kitne prime ulte karne par bhi prime?

11, 13, 17, 31, 37, 71, 73, 79, 97 ⇒ 9 ✓ (b)

① **RRB English • Ch 1 Q13** 1-700 mein 17 se divisible

$\lfloor 700/17 \rfloor = 41$  ✓ (b)

**NS-3 DO NUMBER + SERIES**

**PEHCHAN** sum and difference / योग और अंतर · consecutive numbers / क्रमागत · digits reversed ·  $1 + 2 + \dots + n$  · sum of squares

$$\text{Bada} = \frac{S + D}{2} \quad \text{Chhota} = \frac{S - D}{2}$$

S = dono ka jod · D = dono ka farak

| Sawaal mein ye ho                  | Formula / kya karo                                             | Example: value rakho → answer                         |
|------------------------------------|----------------------------------------------------------------|-------------------------------------------------------|
| ① <b>Jod aur farak diya</b>        | $\text{Bada} = (S + D)/2$                                      | $S = 50, D = 10 \rightarrow 30 \text{ aur } 20$       |
| ② <b>n lagataar numbers ka jod</b> | $\text{Beech wala} = \text{jod} \div n$                        | 5 lagataar, jod 100 → beech 20 → 18...22              |
| ③ <b>2-digit number ulta kiya</b>  | $\text{Farak} = 9 \times (\text{digits ka farak})$             | $72 - 27 = 45 = 9 \times 5$                           |
| ④ <b>1 se n ka jod / squares</b>   | $\frac{n(n+1)}{2} \cdot \text{squares} \frac{n(n+1)(2n+1)}{6}$ | $1 \dots 10 \rightarrow 55 \cdot \text{squares } 385$ |
| ⑤ <b>Tickets / handshakes</b>      | $\text{Tickets} = n(n-1) \cdot \text{handshakes} = n(n-1)/2$   | 5 station → 20 ticket · 10 log → 45 haath             |

**BOOK SE (asli sawaal)**

① **SSC • Number System Q257** Do numbers ka jod 11, squares ka jod 65 ⇒ cubes ka jod?

$S = 11, x^2 + y^2 = 65 \Rightarrow P = (121 - 65)/2 = 28 \Rightarrow x^3 + y^3 = S^3 - 3PS = 1331 - 924 = 407$  ✓ (b)

① **RRB English • Ch 1 Q124** Sum 45, difference 19

$(45 + 19)/2 = 32, 13$  ✓ (b)

## 2 Decimal Fractions दशमलव भिन्न

### DF-1 DECIMAL ↔ FRACTION

**PEHCHAN** p/q form · bar (recurring) decimal · terminating · reciprocal / व्युत्क्रम · lowest form

$$0.\overline{ab} = \frac{ab}{99} \quad 0.a\overline{b} = \frac{ab - a}{90}$$

Repeat digit → 9 · na repeat → 0

| Sawaal mein ye ho            | Formula / kya karo                    | Example: value rakho → answer                          |
|------------------------------|---------------------------------------|--------------------------------------------------------|
| ① Bar wala decimal → p/q     | Repeat ke liye 9, baaki ke liye 0     | $0.36 = 36/99 = 4/11$ · $0.16 = 16/90 = 4/225$         |
| ② Terminating hai?           | Lowest form mein neeche sirf 2 aur 5  | $7/40$ : $40 = 2^3 \times 5 \rightarrow$ haan          |
| ③ Product diya, point khisko | Digits wahi, decimal places gino      | $23 \times 19 = 437 \rightarrow 2.3 \times 1.9 = 4.37$ |
| ④ Reciprocal                 | $\frac{a}{b} \rightarrow \frac{b}{a}$ | $3/7 \rightarrow 7/3$                                  |
| ⑤ Lowest form                | Upar-neeche ÷ HCF                     | $18/24 \rightarrow 3/4$                                |

BOOK SE (asli sawaal)

③ **SSC · Decimal and Fraction Q29**  $x > y > z$ , chhota ÷ bada =  $9/16 = y + 0.0625$ ,  $x + y + z = 1$   $13/24 \Rightarrow x + z?$   
 $y = 9/16 - 0.0625 = 1/2 \Rightarrow x + z = 37/24 - 1/2 = 25/24$  ✓ (c)

① **RRB English · Ch 2 Q73**  $0.\overline{12} = ?$   
 $\frac{12}{99} = \frac{4}{33}$  ✓ (b)

### DF-2 FRACTION: BADA-CHHOTA + JOD

**PEHCHAN** largest / smallest fraction · ascending / descending · kya jodein · bacha hissa · numerator / denominator

$$\frac{a}{b} \text{ vs } \frac{c}{d} : a \times d \text{ vs } b \times c$$

Jiska cross product bada, wahi fraction badi

| Sawaal mein ye ho                         | Formula / kya karo                              | Example: value rakho → answer                      |
|-------------------------------------------|-------------------------------------------------|----------------------------------------------------|
| ① Badi / chhoti fraction                  | Cross guna (ya decimal bana lo)                 | $3/5$ vs $5/8$ : $24$ vs $25 \rightarrow 5/8$ badi |
| ② Kya jodein / bacha hissa                | Kya jodein = $B - A$ · bacha = $1 - \text{jod}$ | $1/2 + 1/3$ diya → bacha $1/6$                     |
| ③ $1/(1 \cdot 2) + 1/(2 \cdot 3) + \dots$ | $\frac{n}{n+1}$                                 | $\dots + 1/(9 \cdot 10) \rightarrow 9/10$          |
| ④ Upar-neeche badla to fraction           | Options daal ke check karo                      | dono +1 → $2/3$ : $3/5 \rightarrow 4/6 = 2/3$ ✓    |

BOOK SE (asli sawaal)

③ **SSC · Decimal and Fraction Q13**  $1/(1 \times 2) + 1/(1 \times 4) + 1/(2 \times 3) + 1/(4 \times 7) + \dots$  20 term = ?  
do cancel series:  $10/11 + (1/3)(1 - 1/31) = 10/11 + 10/31 = 420/341$  ✓ (d)

① **RRB English · Ch 2 Q9**  $15/16, 24/25, 34/35, 19/20$  mein greatest neeche – upar = 1 sab mein ⇒ sabse bade numbers:  $34/35$  ✓ (c)

### 3 Indices and Surds घातांक तथा करणी

#### IS-1 POWER (GHAAT)

**PEHCHAN** power / exponent / घात · find x in  $a^x \cdot \sqrt{\quad}$  inside  $\sqrt{\quad} \cdot \sqrt{x}$  equation

$$a^m \cdot a^n = a^{m+n} \quad \frac{a^m}{a^n} = a^{m-n} \quad (a^m)^n = a^{mn} \quad a^{-n} = \frac{1}{a^n}$$

Same base banao → powers barabar

| Sawaal mein ye ho                        | Formula / kya karo   | Example: value rakho → answer                                 |
|------------------------------------------|----------------------|---------------------------------------------------------------|
| ① x nikaalo                              | Dono taraf same base | $4^x = 8^3 \rightarrow 2^{2x} = 2^9 \rightarrow x = 4.5$      |
| ② $\sqrt{\quad}$ ke andar $\sqrt{\quad}$ | Andar se bahar khol  | $\sqrt{(10 + \sqrt{(30 + \sqrt{36}))}} = \sqrt{(10 + 6)} = 4$ |
| ③ $\sqrt{x}$ wali equation               | Options daalo        | $\sqrt{x} + x = 20 \rightarrow x = 16$                        |

BOOK SE (asli sawaal)

① **SSC · Indices and Surds Q22**  $21^{\sqrt{x}} + 20^{\sqrt{x}} = 29^{\sqrt{x}} \Rightarrow x = ?$

options daalo:  $\sqrt{x} = 2 \Rightarrow 21^2 + 20^2 = 29^2 \checkmark \Rightarrow x = 4 \checkmark$  (c)

① **RRB English · Ch 3 Q115**  $2^x \cdot 4^{12} \cdot 8^3 = 16^{11}$

$x + 24 + 9 = 44 \Rightarrow x = 11 \checkmark$  (b)

#### IS-2 $\sqrt{\quad}$ ROOT + SURD

**PEHCHAN** square root / वर्गमूल ·  $\sqrt{\quad}$  of decimal · surd / करणी · rationalise · which is greater ( $\sqrt{\quad}$ )

$$\sqrt{a^2b} = a\sqrt{b} \quad (a + \sqrt{b})(a - \sqrt{b}) = a^2 - b$$

Neeche  $\sqrt{\quad}$  ho to ulta sign laga ke guna

| Sawaal mein ye ho                      | Formula / kya karo                                | Example: value rakho → answer                               |
|----------------------------------------|---------------------------------------------------|-------------------------------------------------------------|
| ① Bade number ka $\sqrt{\quad}$        | Last digit + paas ka square                       | $\sqrt{5776}$ : 6 → 4/6 · 57 → 7 → 76                       |
| ② Decimal ka $\sqrt{\quad}$            | Point se 2-2 digit                                | $\sqrt{0.0169} = 0.13$                                      |
| ③ Surd jodo                            | Same $\sqrt{\quad}$ mein todo                     | $\sqrt{48} + \sqrt{27} = 4\sqrt{3} + 3\sqrt{3} = 7\sqrt{3}$ |
| ④ Rationalise                          | Neeche ka sign ulta karke guna                    | $1/(\sqrt{3} + \sqrt{2}) = \sqrt{3} - \sqrt{2}$             |
| ⑤ Kaun bada ( $\sqrt{\quad}$ wale jod) | Jod same ho to paas-paas wale numbers ka jod bada | $\sqrt{10} + \sqrt{8} > \sqrt{11} + \sqrt{7}$               |

BOOK SE (asli sawaal)

⑤ **SSC · Indices and Surds Q64**  $P = 2^{29} \cdot 3^{21} \cdot 5^8$ ,  $Q = 2^{27} \cdot 3^{21} \cdot 5^8$ ,  $R = 2^{26} \cdot 3^{22} \cdot 5^8$ ,  $S = 2^{25} \cdot 3^{22} \cdot 5^9 \Rightarrow$  kram?

$2^{25} \cdot 3^{21} \cdot 5^8$  common:  $P \rightarrow 16$ ,  $Q \rightarrow 4$ ,  $R \rightarrow 6$ ,  $S \rightarrow 15 \Rightarrow P > S > R > Q \checkmark$  (a)

① **RRB English · Ch 3 Q13**  $\sqrt{519841}$

last 1  $\Rightarrow 1/9$ ; 5198:  $72^2 = 5184 \Rightarrow 72$ ;  $5198 < 72 \cdot 73 = 5256 \Rightarrow$  chhota  $\Rightarrow$  **721**  $\checkmark$  (a)

## 4 Simplification सरलीकरण

### SIM-1 BODMAS + “?”

**PEHCHAN** simplify / सरल कीजिए · find ? · approximate value · brackets, of

Bracket → of → ÷ → × → + → -

? akela chhodo, baaki ulta karke doosri taraf

| Sawaal mein ye ho           | Formula / kya karo              | Example: value rakho → answer                 |
|-----------------------------|---------------------------------|-----------------------------------------------|
| ① <b>Lamba sawaal</b>       | Upar wala order                 | $8 + 12 \div 4 \times 2 = 8 + 6 = 14$         |
| ② <b>? nikaalo</b>          | × ka ulta ÷, + ka ulta -        | $? \times 12 = 96 \rightarrow ? = 8$          |
| ③ <b>Approximate</b>        | Round karo, phir solve          | $19.98 \times 5.02 \approx 20 \times 5 = 100$ |
| ④ <b>Fraction ki chain</b>  | Neeche se upar                  | $1 + 1/(1 + 1/2) = 1 + 2/3 = 5/3$             |
| ⑤ <b>99 × 99 jaise guna</b> | $(100 - 1)^2 = 10000 - 200 + 1$ | $99^2 = 9801$                                 |

BOOK SE (asli sawaal)

④ **SSC · Simplification Q195**  $(4^3 + 4) \div [5^2 - (7^2 - 41)] = ?$

BODMAS:  $68 \div (25 - 8) = 4$  ✓ (d)

① **RRB English · Ch 4 Q46**  $3 + [3 \times \{3 - (3 + 3) \div 6\}]$

$(3 + 3) \div 6 = 1 \Rightarrow 3 \times 2 = 6 \Rightarrow 9$  ✓ (b)

### SIM-2 IDENTITY SHORTCUT

**PEHCHAN** same numbers repeat ·  $a^2 - b^2$  ·  $x : y$  diya ·  $\sqrt{(n + \sqrt{(n + \dots)})}$

$$a^2 - b^2 = (a + b)(a - b) \quad (a + b)^2 - (a - b)^2 = 4ab$$

Bada calculation mat karo, identity se cancel

| Sawaal mein ye ho                                     | Formula / kya karo          | Example: value rakho → answer                   |
|-------------------------------------------------------|-----------------------------|-------------------------------------------------|
| ① <b>Same numbers baar-baar</b>                       | Identity lagao              | $(57^2 - 43^2) \div 14 = 100$                   |
| ② <b>x : y diya</b>                                   | $x = a, y = b$ seedha daalo | $x : y = 2 : 3 \rightarrow (x + y)/(y - x) = 5$ |
| ③ <b><math>\sqrt{(N + \sqrt{(N + \dots)})}</math></b> | k jisme $k(k - 1) = N$      | $\sqrt{(12 + \sqrt{(12 + \dots)})} = 4$         |

BOOK SE (asli sawaal)

③ **SSC · Simplification Q205**  $1 + 1/2 + \dots + 1/20 = k \Rightarrow 1/4 + 1/6 + \dots + 1/40 = ?$

$1/4 + 1/6 + \dots + 1/40 = \frac{1}{2}(1/2 + \dots + 1/20) = (k - 1)/2$  ✓ (c)

① **RRB English · Ch 4 Q84**  $(3.7)^3 - 3(3.7)^2(0.7) + 3(3.7)(0.7)^2 - (0.7)^3$

$(3.7 - 0.7)^3 = 27$  ✓ (b)

## 5 LCM & HCF लघुत्तम समापवर्त्य एवं महत्तम समापवर्तक

### LH-1 HCF – LCM

**PEHCHAN** HCF / म.स. · LCM / ल.स. · product of two numbers · ratio and HCF · HCF of fractions

$$\text{HCF} \times \text{LCM} = \text{Pehla} \times \text{Doosra}$$

HCF = common, chhoti power · LCM = sab, badi power

| Sawaal mein ye ho              | Formula / kya karo                                                                                                                             | Example: value rakho → answer                                                      |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| ① <b>HCF / LCM nikaalo</b>     | Prime powers todo                                                                                                                              | $12 = 2^2 \cdot 3$ , $18 = 2 \cdot 3^2 \rightarrow \text{HCF } 6, \text{ LCM } 36$ |
| ② <b>HCF, LCM, ek number</b>   | $\text{Doosra} = \text{HCF} \times \text{LCM} \div \text{pehla}$                                                                               | HCF 6, LCM 72, ek 18 $\rightarrow 24$                                              |
| ③ <b>Ratio a : b aur HCF h</b> | Numbers = ha, hb · LCM = h · a · b                                                                                                             | 3 : 4, HCF 5 $\rightarrow 15, 20$ · LCM 60                                         |
| ④ <b>Fraction ka HCF / LCM</b> | $\text{HCF} = \text{HCF}(\text{upar}) / \text{LCM}(\text{neeche}) \cdot \text{LCM}$<br>$= \text{LCM}(\text{upar}) / \text{HCF}(\text{neeche})$ | $1/2, 3/4 \rightarrow \text{HCF } 1/4, \text{ LCM } 3/2$                           |

**BOOK SE** (asli sawaal)

③ **SSC · LCM and HCF Q62** Ratio 7 : 13, HCF 8  $\Rightarrow$  LCM?  
 $\text{LCM} = h \cdot a \cdot b = 8 \times 7 \times 13 = 728$  ✓ (d)

① **RRB English · Ch 5 Q1** LCM of 98, 28, 112  
 $2 \cdot 7^2, 2^2 \cdot 7, 2^4 \cdot 7 \Rightarrow 2^4 \cdot 7^2 = 784$  ✓ (a)

### LH-2 SAATH KAB / SABSE BADA

**PEHCHAN** least number leaving remainder · greatest number which divides · bells ring together / घंटियाँ · largest tile / container

$$\text{Saath / sabse chhota} \Rightarrow \text{LCM} \quad \text{Sabse bada / naap} \Rightarrow \text{HCF}$$

| Sawaal mein ye ho                         | Formula / kya karo         | Example: value rakho → answer                         |
|-------------------------------------------|----------------------------|-------------------------------------------------------|
| ① <b>Sabse chhota N, har baar shesh R</b> | $\text{LCM} + R$           | 4, 6, 8 se shesh 3 $\rightarrow 24 + 3 = 27$          |
| ② <b>Sabse bada jo bhaag de (shesh r)</b> | $\text{HCF}(a - r, b - r)$ | 43, 91, shesh 7 $\rightarrow \text{HCF}(36, 84) = 12$ |
| ③ <b>Sabse bada naap / tiles</b>          | HCF                        | 45 cm, 60 cm $\rightarrow 15$ cm                      |
| ④ <b>Ghantiyan phir saath</b>             | LCM                        | 6 min, 8 min $\rightarrow 24$ min baad                |

**BOOK SE** (asli sawaal)

① **SSC · LCM and HCF Q11** 12, 16, 18, 20, 25 se shesh 4, aur 7 se kate  $\Rightarrow$  x ka hazaar wala digit?  
 $\text{LCM} = 3600 \Rightarrow 3600k + 4$ , 7 se kate  $\Rightarrow k = 5 \Rightarrow 18004 \Rightarrow$  hazaar ka digit 8 ✓ (d)

① **RRB English · Ch 5 Q27** 9, 8, 10, 12 se rem 3  
 $\text{LCM} = 360 \Rightarrow 363$  ✓ (c)

## 6 Percentage प्रतिशतता

### PC-1 % = HISSA

**PEHCHAN** x% of · what percent / कितना प्रतिशत · pass marks / fail by · income, expenditure, savings · votes / election

$$x\% \text{ of } Y = \frac{x \times Y}{100}$$

Y = poora · x = percent

| Sawaal mein ye ho           | Formula / kya karo                         | Example: value rakho → answer        |
|-----------------------------|--------------------------------------------|--------------------------------------|
| ① x% of / kitna %           | Kitna % = $A \div B \times 100$            | 30 is ?% of 120 → 25%                |
| ② Pass marks                | Max = (mila + kitne se fail) × 100 ÷ pass% | 33% pass, 120 mile, 45 se fail → 500 |
| ③ Income – kharcha – bachat | Income = bachat × 100 ÷ bachat%            | 20% bachat = ₹3000 → ₹15000          |
| ④ Election                  | Jeet ka farak = (W% – L%) × valid          | 60% vs 40%, valid 5000 → 1000        |
| ⑤ Do group ka %             | $\frac{n_1p_1 + n_2p_2}{n_1 + n_2}$        | 60 boys (50%), 40 girls (75%) → 60%  |

BOOK SE (asli sawaal)

③ **SSC · Percentage Q127** 18% kiraya, 40% grocery, baaki ka 55% padhai, bachat ₹4725 ⇒ padhai ka kharcha?  
bacha =  $I \times 0.42 \times 0.45 = 4725 \Rightarrow I = 25000 \Rightarrow \text{education} = 0.55 \times 0.42 \times 25000 = 5775$  ✓ (d)

① **RRB English · Ch 6 Q85** 27.5% of number = 11  
 $\frac{11 \times 100}{27.5} = 40$  ✓ (c)

### PC-2 BADHA – GHATA (MULTIPLIER)

**PEHCHAN** increase / decrease · बढ़ा / घटा · successive / do baar · population / जनसंख्या · consumption / खपत

$$\text{Naya} = \text{Purana} \times \frac{100 \pm x}{100}$$

Badha → + · Ghata → –

| Sawaal mein ye ho           | Formula / kya karo                   | Example: value rakho → answer    |
|-----------------------------|--------------------------------------|----------------------------------|
| ① x% badha / ghata          | $\times (100 \pm x)/100$             | 500, 20% badha → 600             |
| ② Do baar change (a%, b%)   | $a + b + ab/100$                     | +20%, -10% → $20 - 10 - 2 = 8\%$ |
| ③ Population n saal         | $P \left(1 + \frac{r}{100}\right)^n$ | 10000, 10%, 2 saal → 12100       |
| ④ x% zyada → ulta kitna kam | $\frac{x}{100 + x} \times 100$       | 25% zyada → 20% kam              |

BOOK SE (asli sawaal)

① **SSC · Percentage Q111** A = B + 25%, B = C – 20%, C = D + 10% ⇒ kaunsa galat?  
D = 100 ⇒ C 110, B 88, A 110 ⇒ B, A se 20% kam (22% nahi) ⇒ (a) galat ✓ (a)

① **RRB English · Ch 6 Q131** 45% badhne par 725  
 $725 \times \frac{100}{145} = 500$  ✓ (a)

## 7 Profit & Loss लाभ और हानि

### PL-1 CP → SP (PROFIT / LOSS)

**PEHCHAN** cost price / क्रय मूल्य · selling price / विक्रय मूल्य · profit % / loss % · marked price + discount · sold for ₹... more

$$SP = CP \times \frac{100 \pm P}{100}$$

P = profit % (loss ho to -)

| Sawaal mein ye ho              | Formula / kya karo                                    | Example: value rakho → answer                                                            |
|--------------------------------|-------------------------------------------------------|------------------------------------------------------------------------------------------|
| ① CP, SP → P%                  | $(SP - CP) \div CP \times 100$                        | CP 400, SP 500 → 25%                                                                     |
| ② SP diya → CP                 | $CP = SP \times 100 \div (100 \pm P)$                 | SP 660, 10% profit → CP 600                                                              |
| ③ ₹ farak ↔ % farak            | $CP = ₹ \text{farak} \times 100 \div \% \text{farak}$ | 10% loss vs 5% profit, farak ₹60 → CP 400                                                |
| ④ Same SP, x% profit + x% loss | Hamesha loss = $x^2/100$ %                            | 20% → 4% loss                                                                            |
| ⑤ MP + discount + profit       | $MP \div CP = (100 + P) \div (100 - D)$               | P 20%, D 20% → MP = 1.5 × CP                                                             |
| ⑥ Shart wale sawaal            | CP = 100 maan lo                                      | SP double → profit triple: $2x - 100 = 3(x - 100) \rightarrow x = 200 \rightarrow 100\%$ |

BOOK SE (asli sawaal)

② **SSC · Profit and Loss Q175** Kapda ₹800 + silai ₹350, 20% profit → SP?

CP = 800 + 350 = 1150 ⇒ SP =  $1150 \times 120/100 = 1380$  ✓ (b)

① **RRB English · Ch 7 Q1** CP ₹4500, SP ₹3500 ⇒ loss %

$\frac{1000}{4500} \times 100 = 22\frac{2}{9}\%$  ✓ (c)

### PL-2 GINTI · RATE · NAKLI BAAT

**PEHCHAN** SP of x articles = CP of y · a for ₹x, b for ₹y · false weight / गलत बाट · part sold at x%, rest at y%

$$P\% = \frac{y - x}{x} \times 100$$

x = jitni beche · y = jitni ki keemat mili

| Sawaal mein ye ho                               | Formula / kya karo                                                | Example: value rakho → answer                          |
|-------------------------------------------------|-------------------------------------------------------------------|--------------------------------------------------------|
| ① x ka SP = y ka CP                             | $(y - x) \div x \times 100$                                       | SP of 20 = CP of 25 → 25%                              |
| ② n beche, loss = k ka CP (ya profit = k ka SP) | CP = kul SP ÷ (n - k) · profit wale mein<br>SP = kul CP ÷ (n - k) | 21 pot ₹2520, loss = 3 ka CP → CP $2520 \div 18 = 140$ |
| ③ a ₹X mein li, b ₹Y mein bechi                 | $(aY \div bX - 1) \times 100$                                     | 3 for ₹10 li, 2 for ₹10 bechi → 50%                    |
| ④ Nakli baat (900 g ko 1 kg)                    | $(\text{asli} - \text{nakli}) \div \text{nakli} \times 100$       | 1000 vs 900 → 11.11%                                   |
| ⑤ Aadha x%, aadha y%                            | Weighted average                                                  | aadha 20% profit, aadha 10% loss → 5%                  |

**BOOK SE (asli sawaal)**

⑤ **SSC • Profit and Loss Q8** 60 pencil 4 for ₹5 + 60 pencil 2 for ₹3, beche 3 for ₹4 ⇒ P/L %?

CP:  $60 \times 5/4 + 60 \times 3/2 = 165$  · SP:  $120 \times 4/3 = 160 \Rightarrow \text{loss } 5/165 = 3 \frac{1}{33} \% \checkmark \text{ (b)}$

① **RRB English • Ch 7 Q3** 32 ka SP = 38 ka CP

$\frac{6}{32} \times 100 = 18.75\% \checkmark \text{ (c)}$

## 8 Discount बढ़ा / छूट

### DS-1 DISCOUNT

**PEHCHAN** marked price / अंकित मूल्य · discount / छूट · successive discount · buy x get y free

$$SP = MP \times \frac{100 - d}{100}$$

MP = likha daam · d = discount %

| Sawaal mein ye ho      | Formula / kya karo                      | Example: value rakho → answer |
|------------------------|-----------------------------------------|-------------------------------|
| ① MP, discount → SP    | $MP \times (100 - d) \div 100$          | MP 800, 25% → 600             |
| ② Do discount (a%, b%) | $a + b - ab/100$                        | 20%, 10% → 28%                |
| ③ MP, discount, profit | $MP \div CP = (100 + P) \div (100 - D)$ | CP 100, P 20, D 20 → MP 150   |
| ④ Buy x get y free     | $y \div (x + y) \times 100$             | Buy 3 get 1 → 25%             |

**BOOK SE** (asli sawaal)

③ **SSC · Discount Q93** 16% discount ke baad bhi 20% gain, CP ₹11900 ⇒ MP?

$$MP = CP \times (100 + 20) / (100 - 16) = 11900 \times 120 / 84 = 17000 \checkmark \text{ (d)}$$

① **RRB English · Ch 8 Q11** MP ₹2850, 21% discount

$$2850 \times 0.79 = 2251.50 \checkmark \text{ (d)}$$

## 9 Ratio & Proportion अनुपात और समानुपात

### RP-1 “1 HISSA” (RATIO)

**PEHCHAN** divide in ratio / अनुपात में बाँटो · A : B and B : C · x% of A = y% of B · coins / सिक्के · income ratio, expenditure ratio

$$1 \text{ hissa} = \frac{\text{Di hui value}}{\text{uske parts}}$$

$$\text{Har cheez} = \text{uske parts} \times 1 \text{ hissa}$$

| Sawaal mein ye ho                | Formula / kya karo                                                                                      | Example: value rakho → answer                                  |
|----------------------------------|---------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| ① ₹T ko a : b mein               | $T \times a \div (a + b)$ · squares ka jod diya $\Rightarrow k^2 = \text{jod} \div (a^2 + b^2 + \dots)$ | ₹500, 2 : 3 $\rightarrow$ 200, 300                             |
| ② A:B, B:C $\rightarrow$ A:B:C   | Beech wala barabar karo                                                                                 | 2 : 3, 4 : 5 $\rightarrow$ 8 : 12 : 15                         |
| ③ x% of A = y% of B              | A : B = y : x (ulta)                                                                                    | 20% A = 30% B $\rightarrow$ 3 : 2                              |
| ④ Coins                          | 1 hissa = kul ₹ $\div \Sigma(\text{ratio} \times \text{value})$                                         | ₹1, 50p, 25p = 1 : 2 : 4, ₹30 $\rightarrow$ 10, 20, 40         |
| ⑤ Income + kharcha ratio, bachat | $k = \frac{S(d - c)}{ad - bc}$ (bachat barabar)                                                         | income 3:2, kharcha 5:3, bachat ₹1000 $\rightarrow$ 6000, 4000 |

BOOK SE (asli sawaal)

② **SSC · Ratio and Proportion Q18** A : B = 11 : 7, B : C = 5 : 19  $\Rightarrow$  A : B : C?

B barabar (35): 55 : 35 : 133 ✓ (a)

① **RRB English · Ch 9 Q2** ₹1470 ko 3 : 4, Mohan (4)

$1470 \times \frac{4}{7} = 840$  ✓ (d)

### RP-2 CROSS GUNA (PROPORTION)

**PEHCHAN** proportion / समानुपात · mean / third / fourth proportional · number added to make ratio · (x + y) : (x - y)

$$\frac{a}{b} = \frac{c}{d} \iff a \times d = b \times c$$

| Sawaal mein ye ho                                                     | Formula / kya karo                                                         | Example: value rakho → answer                     |
|-----------------------------------------------------------------------|----------------------------------------------------------------------------|---------------------------------------------------|
| ① Kuch joda $\rightarrow$ naya ratio                                  | Cross guna                                                                 | 5:7, dono mein 2 jodo $\rightarrow$ 7:9           |
| ② 4th / 3rd / mean                                                    | $4\text{th} = bc/a \cdot 3\text{rd} = b^2/a \cdot \text{mean} = \sqrt{ab}$ | 4, 6, 9 ka 4th = $13.5 \cdot 4$ aur 9 ka mean = 6 |
| ③ $(x + y) \div (x - y) = p \div q$ $x \div y = (p + q) \div (p - q)$ |                                                                            | $5/3 \rightarrow x/y = 8/2 = 4$                   |

BOOK SE (asli sawaal)

① **SSC · Ratio and Proportion Q100** 52, 47, 20, 19 se x ghatane par proportion  $\Rightarrow (x + 13), (x - 8)$  ka mean proportional?

cross guna:  $(52 - x)(19 - x) = (47 - x)(20 - x) \Rightarrow x = 12 \Rightarrow$  mean of 25, 4 = 10 ✓ (b)

① **RRB English · Ch 9 Q79** 2 : 3, dono mein 4  $\Rightarrow$  7 : 10; farak

$k = \frac{4(7-10)}{20-21} = 12 \Rightarrow 12$  ✓ (c)

## 10 Partnership साझेदारी

### PT-1 PROFIT = PAISA × SAMAY

**PEHCHAN** partnership / साझेदारी · invested / निवेश · share of profit · joined after ... months

$$A : B = \text{Paisa}_A \times \text{Mahine}_A : \text{Paisa}_B \times \text{Mahine}_B$$

| Sawaal mein ye ho      | Formula / kya karo       | Example: value rakho → answer               |
|------------------------|--------------------------|---------------------------------------------|
| ① Profit baanto        | Paisa × mahine ka ratio  | A 5000 × 12, B 6000 × 10 → 1 : 1            |
| ② Anjaan paisa / samay | Ratio ulta karke nikaalo | A 4000 (12 m), B 8 m, profit 2:3 → B = 9000 |

**BOOK SE** (asli sawaal)

① **SSC · Partnership Q20** ₹24000 (8 mahine), ₹28000 (10), ₹20000 (12), profit ₹25810 ⇒ pehle ka hissa?

$$24000 \times 8 : 28000 \times 10 : 20000 \times 12 = 24 : 35 : 30 \Rightarrow 25810 \times 24/89 = 6960 \checkmark \text{ (a)}$$

① **RRB English · Ch 10 Q26** ₹4000, ₹8000, ₹6000; Sharma 6 mahine baad gaya; 8 mahine ka ₹34,000

$$24 : 64 : 48 \Rightarrow 34000 \times \frac{64}{136} = 16000 \checkmark \text{ (d)}$$

## 11 Work & Time समय और कार्य

### WT-1 KAAM = LCM

**PEHCHAN** work / काम · together / साथ मिलकर · alone / अकेले · efficient · left after / beech mein chhoda · wages / मजदूरी

$$\text{Kul kaam} = \text{LCM}(\text{din}) \quad \text{Saath ke din} = \frac{a \times b}{a + b}$$

a, b = akele ke din

| Sawaal mein ye ho          | Formula / kya karo                                                                                                                     | Example: value rakho → answer                            |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| ① Saath / akela            | $ab \div (a + b)$                                                                                                                      | A 10 din, B 15 din → 6 din                               |
| ② A+B, B+C, C+A diye       | Teeno = $2 \div (1/x + 1/y + 1/z)$                                                                                                     | 12, 15, 20 → 10 din                                      |
| ③ k guna efficient         | Dheema = $T(k + 1) \cdot \text{tez} = T(k + 1) \div k$                                                                                 | A = 2B, saath 10 din → B 30, A 15                        |
| ④ Beech mein chhoda / juda | $\Sigma (\text{kaam kiye din} \div \text{akele ke din}) = 1 \cdot A, k$<br>din pehle chhode ⇒ kul din = $(a + k) \cdot b \div (a + b)$ | A(10) ne 4 din kiya, baaki B(15) → 9 din                 |
| ⑤ Baari-baari              | 1 cycle ka kaam jodo                                                                                                                   | A 4, B 6 din, baari-baari (A pehle) → $4\frac{2}{3}$ din |
| ⑥ Mazdoori                 | Paisa $\propto$ kaam (din ka ulta)                                                                                                     | A 10, B 15 din, ₹500 → 300, 200                          |

BOOK SE (asli sawaal)

- ③ **SSC · Time and Work Q234** A = B se 50% tez, C = B se 40% dheema, saath 10 din ⇒ A akela 150% kaam?  
 $A : B : C = 15 : 10 : 6 \Rightarrow \text{kaam} = 31 \times 10 = 310 \Rightarrow 150\% = 465 \Rightarrow 465 \div 15 = 31 \checkmark \text{ (c)}$
- ① **RRB English · Ch 11 Q5** A 15, B 20 ⇒ saath  
 $\frac{15 \times 20}{35} = 8\frac{4}{7} \text{ din } \checkmark \text{ (c)}$

### WT-2 LOG × DIN × GHANTE

**PEHCHAN** men, women / पुरुष, महिलाएँ · days, hours per day · food lasts / भोजन

$$\frac{M_1 D_1 H_1}{W_1} = \frac{M_2 D_2 H_2}{W_2}$$

M = log · D = din · H = ghante · W = kaam

| Sawaal mein ye ho  | Formula / kya karo           | Example: value rakho → answer    |
|--------------------|------------------------------|----------------------------------|
| ① Log, din, ghante | Jo poocha use x, baaki daalo | 10 log 12 din → 15 log ? → 8 din |

BOOK SE (asli sawaal)

- ① **SSC · Time and Work Q69** 12 din, 10 ghante roz ⇒ 8 din mein roz kitne ghante?  
 $1 \times 12 \times 10 = 1 \times 8 \times x \Rightarrow x = 15 \checkmark \text{ (a)}$
- ① **RRB English · Ch 11 Q70** 12 men 24.5 din 1 wall ⇒ 5 walls 49 din  
 $\frac{12 \times 24.5 \times 5}{49} = 30 \checkmark \text{ (b)}$

## 12 Alligation / Mixture मिश्रण / एलीगेशन

### MX-1 ALLIGATION (CROSS)

**PEHCHAN** in what ratio mixed / किस अनुपात में · mixture price · mean price

$$\frac{\text{Sasta}}{\text{Mehenga}} = \frac{d - m}{m - c}$$

$$c = \text{sasta} \cdot d = \text{mehenga} \cdot m = \text{mix ka daam}$$

| Sawaal mein ye ho                 | Formula / kya karo                                             | Example: value rakho → answer   |
|-----------------------------------|----------------------------------------------------------------|---------------------------------|
| ① <b>Kis ratio mein milaayein</b> | Cross: $(d - m) : (m - c)$                                     | ₹20 + ₹30 → ₹26 → 4 : 6 = 2 : 3 |
| ② <b>Mix ka daam</b>              | $\Sigma(\text{qty} \times \text{daam}) \div \Sigma \text{qty}$ | 2 kg ₹20 + 3 kg ₹30 → ₹26       |

**BOOK SE (asli sawaal)**

② **SSC • Alligation Q14** Do total 35% aur  $33\frac{1}{3}\%$  orange, baaki apple, mila diya ⇒ apple %?  
apple 65% aur  $66\frac{2}{3}\%$ , barabar quantity ⇒ average =  $65\frac{5}{6}\%$  ✓ (b)

① **RRB English • Ch 12 Q10** 80% juice kitna milayein 36 L ke 25% mein ⇒ 60%  
 $25 : 80 \Rightarrow (80 - 60) : (60 - 25) = 20 : 35 \Rightarrow 36 \times \frac{35}{20} = 63$  ✓ (a)

### MX-2 PAANI MILAO / NIKAALO

**PEHCHAN** water added / पानी मिलाया · evaporated · taken out and replaced

$$C_1 V_1 = C_2 V_2 \quad \text{Bacha} = \text{Shuru} \times \left(1 - \frac{x}{V}\right)^n$$

Jo cheez nahi badli, wo fix

| Sawaal mein ye ho           | Formula / kya karo  | Example: value rakho → answer                         |
|-----------------------------|---------------------|-------------------------------------------------------|
| ① <b>Paani joda / udaya</b> | $C_1 V_1 = C_2 V_2$ | 40 L, 10% namak → 8%: $V = 50 \rightarrow 10$ L paani |
| ② <b>Baar-baar nikaalo</b>  | $(1 - x/V)^n$       | 40 L doodh, 4 L, 2 baar → 32.4                        |

**BOOK SE (asli sawaal)**

① **SSC • Alligation Q16** Juice : paani = 5 : x, 4 L mein 1 L paani ⇒ 1 : 1 ⇒ x?  
options daalo:  $x = 3 \Rightarrow$  juice 2.5, paani  $1.5 + 1 = 2.5 \Rightarrow 1 : 1$  ✓✓ (a)

① **RRB English • Ch 12 Q1** 90 ml 38% ⇒ 17.1% ke liye paani  
 $V_2 = \frac{90 \times 38}{17.1} = 200 \Rightarrow 110$  ml ✓ (c)

## 13 Pipe & Cistern नल और टंकी

### PP-1 NAL – TANKI

**PEHCHAN** pipe, tap / नल · tank / टंकी · fill + empty · leak / रिसाव · opened alternately

$$\frac{1}{T} = \frac{1}{a} + \frac{1}{b} - \frac{1}{c}$$

a, b = bharne wale · c = khaali karne wala

| Sawaal mein ye ho                     | Formula / kya karo                      | Example: value rakho → answer             |
|---------------------------------------|-----------------------------------------|-------------------------------------------|
| ① Kai nal saath                       | Bharo +, khaali –                       | A 10 h, B 15 h → 6 h                      |
| ② Leak                                | Leak = $ab \div (b - a)$                | 6 h ki jagah 8 h → leak 24 h              |
| ③ Saath T; A ko a zyada, B ko b zyada | $T = \sqrt{ab}$                         | a = 4, b = 9 → 6 h                        |
| ④ Beech mein band / khola             | $\Sigma$ (khula samay ÷ akela time) = 1 | A 12, B 18 saath 4 h, phir A → 5 h 20 min |
| ⑤ Baari-baari ghante                  | 2 ghante ka net, aakhri ghante dhyan    | bharo 6 h, khaali 12 h, alternate → 21 h  |
| ⑥ Capacity (litre)                    | $C = r \div  1/T - 1/a $                | 5 h, leak 10 L/h ke saath 6 h → 300 L     |

BOOK SE (asli sawaal)

① **SSC · Pipes and Cisterns Q8** A, B, C saath 4 h; C = 2B, B = 3A ⇒ A akela?

A : B : C = 1 : 3 : 6 ⇒ kaam =  $10 \times 4 = 40 \Rightarrow$  A akela 40 h ✓ (b)

① **RRB English · Ch 13 Q1** A 21 h, B 18 h ⇒ saath

$\frac{21 \times 18}{39} = 9\frac{9}{13}$  h ✓ (a)

## 14 Simple Interest साधारण ब्याज

### SI-1 SIMPLE INTEREST (P·R·T)

**PEHCHAN** simple interest / साधारण ब्याज · amount / मिश्रधन · rate, time · becomes double / n times

$$SI = \frac{P \times R \times T}{100} \quad A = P + SI$$

$P = \text{mool} \cdot R = \text{dar \%} \cdot T = \text{saal}$

| Sawaal mein ye ho                                                     | Formula / kya karo                                             | Example: value rakho → answer                |
|-----------------------------------------------------------------------|----------------------------------------------------------------|----------------------------------------------|
| ① P, R, T → SI                                                        | $PRT \div 100$                                                 | 5000, 10%, 2 saal → 1000                     |
| ② T <sub>1</sub> pe A <sub>1</sub> , T <sub>2</sub> pe A <sub>2</sub> | $1 \text{ saal SI} = (A_2 - A_1) \div (T_2 - T_1)$             | 2 saal 1200, 5 saal 1500 → 100/saal → P 1000 |
| ③ Rate badla, farak diya                                              | $P = \text{farak} \times 100 \div (\Delta R \times T)$         | rate +2%, 3 saal, farak ₹60 → P 1000         |
| ④ n guna                                                              | $R \times T = (n - 1) \times 100$                              | double 10 saal mein → R = 10%                |
| ⑤ Do hisse, SI barabar                                                | $P_1 : P_2 = 1/(R_1 T_1) : 1/(R_2 T_2)$                        | 5%·2 saal vs 4%·5 saal → 2 : 1               |
| ⑥ Kisht (instalment)                                                  | $\text{Kisht} = 100 \times D \div [100n + R \cdot n(n - 1)/2]$ | ₹1030, 2 kisht, 6% → ₹500                    |

**BOOK SE** (asli sawaal)

② **SSC · Simple Interest Q71** SI: 5 saal mein ₹500, 7 saal mein ₹600 ⇒ mool?

1 saal SI =  $(600 - 500)/2 = 50 \Rightarrow P = 500 - 5 \times 50 = 250$  ✓ (d)

① **RRB English · Ch 14 Q2** ₹48,750, 16%, 73 din

$\frac{48750 \times 16 \times 73}{36500} = 1560$  ✓ (a)

## 15 Compound Interest चक्रवृद्धि ब्याज

### CI-1 COMPOUND INTEREST

**PEHCHAN** compound interest / चक्रवृद्धि · compounded half-yearly / quarterly · amount after n years · instalments

$$A = P \left( 1 + \frac{R}{100} \right)^n$$

Half-yearly:  $R/2$ ,  $2n$  · Quarterly:  $R/4$ ,  $4n$  ·  $CI = A - P$

| Sawaal mein ye ho                    | Formula / kya karo                                                                                                        | Example: value rakho → answer     |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| ① <b>A / CI nikaalo</b>              | $P(1 + R/100)^n$ · adhoora saal $2\frac{3}{4}$ : $P(1 + r)^2 \times (1 + \frac{3}{4}r)$ , $r = R/100$                     | 1000, 10%, 2 saal → 1210 (CI 210) |
| ② <b>Rate / do amount diye</b>       | $(A/P)$ ka nth root - 1 · lagataar saal: $R = (A_2 - A_1) \div A_1 \times 100 \cdot (1 + r)^{(n_2 - n_1)} = A_2 \div A_1$ | 1000 → 1331, 3 saal → 10%         |
| ③ <b>k guna</b>                      | 2 guna T saal → 4 guna 2T saal                                                                                            | 2 guna 5 saal → 8 guna 15 saal    |
| ④ <b>Kisht</b>                       | $P = x/(1 + r) + x/(1 + r)^2$                                                                                             | 2 kisht ₹441, 5% → P 820          |
| ⑤ <b>Half-yearly vs yearly farak</b> | $P[(1 + R/200)^{2n} - (1 + R/100)^n]$                                                                                     | 10000, 10%, 1 saal → 25           |

**BOOK SE** (asli sawaal)

① **SSC · Compound Interest Q11** CI: 3 saal mein ₹8028, 6 saal mein ₹12042 ⇒ mool?

$(1 + R)^3 = 12042/8028 = 1.5 \Rightarrow P = 8028 \div 1.5 = 5352$  ✓ (a)

① **RRB English · Ch 15 Q1** 10% pe 2 saal mein ₹10,164 ⇒ P

$\frac{10164}{1.21} = 8400$  ✓ (b)

### CI-2 CI – SI FARAK

**PEHCHAN** difference between CI and SI · CI and SI for 2 years

$$2 \text{ saal: } P \left( \frac{R}{100} \right)^2 \quad 3 \text{ saal: } P \left( \frac{R}{100} \right)^2 \left( 3 + \frac{R}{100} \right)$$

| Sawaal mein ye ho     | Formula / kya karo      | Example: value rakho → answer |
|-----------------------|-------------------------|-------------------------------|
| ① <b>Farak 2 saal</b> | $P(R/100)^2$            | 10000, 10% → 100              |
| ② <b>Farak 3 saal</b> | $P(R/100)^2(3 + R/100)$ | 10000, 10% → 310              |

**BOOK SE** (asli sawaal)

① **SSC · Compound Interest Q83** 11% par 2 saal ka CI – SI = ₹60.50 ⇒ mool?

2 saal:  $D = P(R/100)^2 \Rightarrow P = 60.50 \div 0.11^2 = 5000$  ✓ (d)

① **RRB English · Ch 15 Q132** 2 saal 9%, farak ₹97.2 ⇒ P

$\frac{97.2 \times 10^4}{81} = 12000$  ✓ (d)

## 16 Problems on Age आयु संबंधी प्रश्न

### AG-1 UMAR (AGE)

**PEHCHAN** age / आयु · years ago / वर्ष पहले · after ... years / वर्ष बाद · ratio of ages

$$\text{Umar} = ax, bx \quad \frac{ax + t}{bx + t} = \frac{c}{d} \Rightarrow x = \frac{t(c - d)}{ad - bc}$$

t saal pehle ho to t minus

| Sawaal mein ye ho                      | Formula / kya karo               | Example: value rakho → answer                                           |
|----------------------------------------|----------------------------------|-------------------------------------------------------------------------|
| ① <b>Abhi a : b, t saal baad c : d</b> | $x = t(c - d) \div (ad - bc)$    | 3 : 4, 5 saal baad 4 : 5 → x = 5 → 15, 20                               |
| ② <b>Farak</b>                         | Farak kabhi nahi badalta         | A 10 saal bada, ratio 2 : 1 → 20, 10                                    |
| ③ <b>Jod / average</b>                 | n logon ka jod t saal baad + n·t | 3 log ka jod 60, 5 saal baad → 75                                       |
| ④ <b>Baaki sawaal</b>                  | Options daal ke check            | Pita = 3 × beta, 10 saal baad 2 guna → beta 10, pita 30 (40 = 2 × 20 ✓) |

**BOOK SE** (asli sawaal)

① **SSC • Age Problems Q20** 7 saal pehle 4 : 5, 7 saal baad 5 : 6 ⇒ 5 saal baad ratio?

t = 14:  $x = 14(5 - 6)/(4 \cdot 6 - 5 \cdot 5) = 14 \Rightarrow 56, 70 \Rightarrow 5 \text{ saal baad } 68 : 82 = 34 : 41 \checkmark \text{ (a)}$

① **RRB English • Ch 16 Q5** abhi 3 guna, 10 saal baad 2 guna ⇒ beti

$k = \frac{10(2-1)}{3-2} = 10 \checkmark \text{ (d)}$

## 17 Average औसत

### AV-1 AVERAGE = TOTAL ÷ n

**PEHCHAN** average / औसत · joins / leaves / replaced · runs / marks · consecutive numbers

$$\text{Total} = \text{Average} \times n$$

n = kitne log / numbers

| Sawaal mein ye ho      | Formula / kya karo                                                                                    | Example: value rakho → answer                          |
|------------------------|-------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| ① Lagataar numbers     | (pehla + aakhri) ÷ 2                                                                                  | 1 se 9 → 5                                             |
| ② Missing number       | n × avg – baaki ka jod                                                                                | 5 ka avg 20, 4 ka jod 75 → 25                          |
| ③ Do group             | (n <sub>1</sub> a <sub>1</sub> + n <sub>2</sub> a <sub>2</sub> ) ÷ (n <sub>1</sub> + n <sub>2</sub> ) | 20 (avg 50) + 30 (avg 60) → 56                         |
| ④ Ek badla / naya aaya | Badla: naya = purana + n × d · Aaya: naya aadmi = purana avg + (n + 1) × d (d = avg ka badlav)        | 10 log, avg +2, 40 kg gaya → naya 60 kg                |
| ⑤ Pehle k + aakhri k   | Beech wala = (dono hisson ka jod) – kul                                                               | 11 ka avg 50, pehle 6 ka 49, aakhri 6 ka 52 → 6th = 56 |
| ⑥ Kitne runs chahiye   | Target × kul – abhi tak                                                                               | 10 match avg 40 chahiye, 9 mein 350 → 50               |

**BOOK SE (asli sawaal)**

② **SSC · Number System Q244** 1088 numbers ka average 0 ⇒ zyada se zyada kitne negative?

Total = avg × n = 0 ⇒ ek positive baaki sab ko balance kar sakta ⇒ 1088 – 1 = 1087 ✓ (b)

① **RRB English · Ch 17 Q2** 11 ke pehle 12 multiples

$$\frac{11 \times 13}{2} = 71.5 \text{ ✓ (c)}$$

## 18 Speed, Time & Distance चाल, समय और दूरी

### ST-1 DOORI = SPEED × SAMAY

**PEHCHAN** speed, distance, time / चाल, दूरी, समय · km/h, m/s · late / early / देर, जल्दी · usual speed ka 3/4

$$D = S \times T \quad \text{km/h} \times \frac{5}{18} = \text{m/s}$$

D = doori · S = speed · T = samay

| Sawaal mein ye ho           | Formula / kya karo                                                                          | Example: value rakho → answer                     |
|-----------------------------|---------------------------------------------------------------------------------------------|---------------------------------------------------|
| ① D, S, T                   | $D = S \times T$                                                                            | 60 km/h, 2.5 h → 150 km                           |
| ② Late / early (do speeds)  | $D = \frac{S_1 S_2}{S_2 - S_1} \times \Delta t$                                             | 4 km/h → 10 min late, 5 km/h → 5 min jaldi → 5 km |
| ③ Speed ka ratio            | Same D: speed ulta samay                                                                    | 3/4 speed se 20 min late → usual 60 min           |
| ④ Stoppage (rukne ka samay) | Har ghante ruke = (speed ka farak ÷ bina ruke speed) × 60 min                               | 80 bina ruke, 72 ruk ke → 6 min                   |
| ⑤ Do sawari (bus + car)     | Tez ka samay = $(D - S_{\text{dheema}} \times T) \div (S_{\text{tez}} - S_{\text{dheema}})$ | 100 km, 3 h, bus 30, car 40 → car 1 h             |

**BOOK SE** (asli sawaal)

① **SSC · Train Q19** Train = car ki  $2\frac{1}{6}$  guna speed, car 486 km 9 h mein ⇒ train 6 h mein?  
car  $486 \div 9 = 54 \Rightarrow$  train  $54 \times 13/6 = 117 \Rightarrow$  6 h mein 702 km ✓ (b)

① **RRB English · Ch 18 Q3** 57.6 km in 48 min ⇒ m/s  
 $\frac{57.6}{0.8} = 72 \text{ km/h} \times \frac{5}{18} = 20$  ✓ (d)

### ST-2 AVERAGE + RELATIVE SPEED

**PEHCHAN** average speed / औसत चाल · towards each other / आमने-सामने · same direction / पीछा · meet / overtake

$$\text{Avg} = \frac{2xy}{x+y} \quad \text{Relative} = S_1 + S_2 \text{ (aamne)}, S_1 - S_2 \text{ (peeche)}$$

x, y = jaane / aane ki speed

| Sawaal mein ye ho          | Formula / kya karo                                  | Example: value rakho → answer        |
|----------------------------|-----------------------------------------------------|--------------------------------------|
| ① Average speed            | Kul doori ÷ kul samay · aana-jaana $2xy \div (x+y)$ | 40 jaana, 60 aana → 48               |
| ② Milna / pakadna          | Gap ÷ $(S_1 + S_2)$ · Gap ÷ $(S_1 - S_2)$           | 100 km, 30 aur 20 aamne-saamne → 2 h |
| ③ Milne ke baad $t_1, t_2$ | $S_1 : S_2 = \sqrt{t_2} : \sqrt{t_1}$               | 4 h aur 9 h → 3 : 2                  |

**BOOK SE** (asli sawaal)

① **SSC · Speed, Time and Distance Q149** Avg 80 km/h, 75% doori  $\frac{2}{3}$  samay mein ⇒ baaki ki speed?  
 $D = 80, T = 1: 25\% = 20 \text{ km in } \frac{1}{3} \text{ h} \Rightarrow 60 \text{ km/h}$  ✓ (c)

① **RRB English · Ch 18 Q112** 45 jaana, 36 aana  
 $\frac{2 \cdot 45 \cdot 36}{81} = 40$  ✓ (c)

## 19 Train रेलगाड़ी

### TR-1 TRAIN

**PEHCHAN** train / रेलगाड़ी · pole, man, tree / खंभा · platform, bridge, tunnel · two trains / दो ट्रेनें

$$\text{Samay} = \frac{L_{\text{train}} + L_{\text{cheez}}}{\text{Relative speed}}$$

$$\text{Pole / aadmi ki lambai} = 0 \cdot \text{km/h} \times 5/18 = \text{m/s}$$

| Sawaal mein ye ho         | Formula / kya karo                                          | Example: value rakho → answer                |
|---------------------------|-------------------------------------------------------------|----------------------------------------------|
| ① Pole / aadmi            | $t = L \div S$                                              | 200 m, 72 km/h (20 m/s) → 10 s               |
| ② Platform / pul          | $t = (L + P) \div S$                                        | 200 + 300 m, 20 m/s → 25 s                   |
| ③ Do platform (alag time) | $S = (P_2 - P_1) \div (t_2 - t_1)$                          | 100 m in 15 s, 250 m in 20 s → 30 m/s, L 350 |
| ④ Do trains               | $(L_1 + L_2) \div (S_1 + S_2)$ aamne · $(S_1 - S_2)$ peeche | 150 + 250 m, 25 aur 15 m/s aamne → 10 s      |
| ⑤ Chalta aadmi            | $t = L \div (S \mp s)$                                      | 100 m, 12 m/s, aadmi 2 m/s same → 10 s       |
| ⑥ Bina lambai milna       | $\text{Gap} \div (S_1 \pm S_2)$                             | 300 km, 60 aur 40 aamne → 3 h                |

**BOOK SE** (asli sawaal)

③ **SSC · Train Q65** Train pole ko 1 min mein; dugni lambai ka pul kitne second mein?

$$L = 60S \Rightarrow (L + 2L) \div S = 180 \text{ s} \checkmark \text{ (d)}$$

① **RRB English · Ch 19 Q21** 72 km/h, pole 12 s ⇒ lambai

$$20 \times 12 = 240 \text{ m} \checkmark \text{ (d)}$$

## 20 Boat & Stream नाव और धारा

### BS-1 NAAV – DHAARA

**PEHCHAN** boat, stream / नाव, धारा · downstream / upstream · still water / शांत जल

$$B = \frac{D + U}{2} \quad S = \frac{D - U}{2}$$

D = dhaara ke saath speed · U = ulti speed · B = naav · S = dhaara

| Sawaal mein ye ho               | Formula / kya karo                             | Example: value rakho → answer   |
|---------------------------------|------------------------------------------------|---------------------------------|
| ① D, U se B, S                  | Upar wala formula                              | D 12, U 8 → B 10, S 2           |
| ② Ulta samay = n × seedha samay | B : S = (n + 1) : (n - 1)                      | n = 2 → 3 : 1                   |
| ③ Jaana + aana kul T            | d = T(B <sup>2</sup> - S <sup>2</sup> ) ÷ (2B) | B 10, S 2, T 5 h → 24 km        |
| ④ Do trip, do samay             | Options ka B daal ke check                     | B 10, S 2: 24/12 + 16/8 = 4 h ✓ |

BOOK SE (asli sawaal)

① **SSC · Boat and Stream Q16** Ulta 20 km/h, seedha 30 km/h ⇒ shaant paani mein naav?

$$B = (D + U)/2 = (30 + 20)/2 = 25 \text{ ✓ (c)}$$

① **RRB English · Ch 20 Q2** B 14; 28 km downstream 1<sup>3</sup>/<sub>4</sub> h

$$D = 16 \Rightarrow S = 2 \text{ ✓ (a)}$$

## 21 Mensuration क्षेत्रमिति

### MN-1 2-D AREA TABLE

**PEHCHAN** area / क्षेत्रफल · perimeter / परिमाप · triangle, rectangle, square, rhombus · circle, wheel, path · hexagon

| Shape       | Area                             | Perimeter   |
|-------------|----------------------------------|-------------|
| Triangle    | $\frac{1}{2}bh$                  | $a + b + c$ |
| Equilateral | $\frac{\sqrt{3}}{4}a^2$          | $3a$        |
| Rectangle   | $l \times b$                     | $2(l + b)$  |
| Square      | $a^2$                            | $4a$        |
| Rhombus     | $\frac{1}{2}d_1d_2$              | $4a$        |
| Trapezium   | $\frac{1}{2}(a + b)h$            |             |
| Circle      | $\pi r^2$                        | $2\pi r$    |
| Hexagon     | $6 \times \frac{\sqrt{3}}{4}a^2$ | $6a$        |

Heron:  $\sqrt{s(s-a)(s-b)(s-c)}$ ,  $s = (a+b+c)/2$

| Sawaal mein ye ho                    | Formula / kya karo                                            | Example: value rakho → answer               |
|--------------------------------------|---------------------------------------------------------------|---------------------------------------------|
| ① Triangle                           | $\frac{1}{2}bh$ · equilateral $\frac{\sqrt{3}}{4}a^2$ · Heron | sides 13, 14, 15 → s 21 → 84                |
| ② Right triangle: perimeter P, hyp h | Area = $[(P - h)^2 - h^2] \div 4$                             | P 60, h 26 → 120                            |
| ③ Rectangle / square / rhombus       | Table · rhombus side <sup>2</sup> = $(d_1/2)^2 + (d_2/2)^2$   | rhombus d 6, 8 → area 24, side 5            |
| ④ Path / border                      | Bahar path: $(l + 2w)(b + 2w) - lb$                           | $10 \times 8$ , path 1 m → $40 \text{ m}^2$ |
| ⑤ Circle / wheel                     | $2\pi r$ , $\pi r^2$ · chakkar = doori ÷ $2\pi r$             | $r = 7 \rightarrow 44, 154$                 |
| ⑥ Hexagon                            | $6 \times \frac{\sqrt{3}}{4} \times a^2$                      | $a = 2 \rightarrow 6\sqrt{3}$               |

BOOK SE (asli sawaal)

③ **SSC · Mensuration 2D Q164** Chaudai =  $4 \times \text{lambai}$ , area 1764 ⇒ chaudai?  
 $x \times 4x = 1764 \Rightarrow x = 21 \Rightarrow \text{chaudai } 84 \checkmark \text{ (b)}$

① **RRB English · Ch 21 Q3** hyp 13, ek side 5  
doosri  $12 \Rightarrow \frac{1}{2} \cdot 5 \cdot 12 = 30 \checkmark \text{ (c)}$

**MN-2 3-D VOLUME TABLE**

**PEHCHAN** volume / आयतन · surface area / पृष्ठ · cube, cuboid, cylinder · cone, sphere, hemisphere · melted and recast / पिघलाकर · prism, pyramid

| Shape      | Volume                        | Surface (TSA)     |
|------------|-------------------------------|-------------------|
| Cube       | $a^3$                         | $6a^2$            |
| Cuboid     | $l b h$                       | $2(lb + bh + hl)$ |
| Cylinder   | $\pi r^2 h$                   | $2\pi r(r + h)$   |
| Cone       | $\frac{1}{3}\pi r^2 h$        | $\pi r(r + l)$    |
| Sphere     | $\frac{4}{3}\pi r^3$          | $4\pi r^2$        |
| Hemisphere | $\frac{3}{8}\pi r^3$          | $3\pi r^2$        |
| Prism      | Base $\times h$               |                   |
| Pyramid    | $\frac{1}{3}$ Base $\times h$ |                   |

Cone:  $l = \sqrt{(r^2 + h^2)}$  · curved: cylinder  $2\pi rh$ , cone  $\pi rl$ , sphere  $4\pi r^2$

| Sawaal mein ye ho              | Formula / kya karo                                                                                     | Example: value rakho → answer                                     |
|--------------------------------|--------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| ① <b>Cube / cuboid</b>         | Table · diagonal: $a\sqrt{3}$ , $\sqrt{(l^2 + b^2 + h^2)}$ · TSA = $(l + b + h)^2 - \text{diagonal}^2$ | cube $a = 3 \rightarrow V 27$ , TSA 54                            |
| ② <b>Cylinder</b>              | $\pi r^2 h \cdot 2\pi r(r + h)$                                                                        | $r 7, h 10 \rightarrow V 1540$                                    |
| ③ <b>Cone</b>                  | $l = \sqrt{(r^2 + h^2)}$                                                                               | $r 3, h 4 \rightarrow l 5, V 12\pi$                               |
| ④ <b>Sphere / hemisphere</b>   | $4/3 \pi r^3$                                                                                          | $r 3 \rightarrow 36\pi$                                           |
| ⑤ <b>Pighla ke naya banaya</b> | Volume same $\rightarrow n = V \text{ bada} \div V \text{ chhota}$                                     | $r 6 \text{ ka gola} \rightarrow r 2 \text{ ke } 27 \text{ gole}$ |
| ⑥ <b>Prism / pyramid</b>       | Prism = base $\times h$ · pyramid = $\frac{1}{3} \times \text{base} \times h$                          | base 12, $h 5 \rightarrow \text{prism } 60$ , pyramid 20          |

**BOOK SE (asli sawaal)**

⑥ **SSC · Mensuration 3D Q194** Pyramid ka  $V = 45\sqrt{3}$ , base equilateral side 6  $\Rightarrow$  height?

$$\frac{1}{3} \times (\sqrt{3}/4 \times 36) \times h = 45\sqrt{3} \Rightarrow 3\sqrt{3} h = 45\sqrt{3} \Rightarrow h = 15 \checkmark \text{ (b)}$$

① **RRB English · Ch 21 Q229** painting ₹588 @ ₹2/cm<sup>2</sup>

$$\text{TSA } 294 \Rightarrow a 7 \Rightarrow 343 \checkmark \text{ (a)}$$

**MN-3 SIZE BADLA**

**PEHCHAN** radius doubled · side increased by  $x\%$  · ratio of volumes / areas

$$\text{Lambai} \times k \Rightarrow \text{Area} \times k^2, \quad \text{Volume} \times k^3$$

| Sawaal mein ye ho                      | Formula / kya karo        | Example: value rakho → answer                                 |
|----------------------------------------|---------------------------|---------------------------------------------------------------|
| ① <b>Lambai k guna</b>                 | Area $k^2$ , volume $k^3$ | radius double $\rightarrow$ area $4\times$ , volume $8\times$ |
| ② <b>Lambai <math>x\%</math> badhi</b> | Area: $2x + x^2/100$      | $+10\% \rightarrow$ area $+21\%$                              |

**BOOK SE (asli sawaal)**

① **SSC · Mensuration 3D Q19** Do cube ka volume 64 : 27  $\Rightarrow$  edges ka ratio?

$$\text{volume } a^3 : b^3 = 64 : 27 \Rightarrow \text{side } 4 : 3 \checkmark \text{ (a)}$$

① **RRB English · Ch 21 Q86** radius  $+5\%$

$$10 + 0.25 = 10.25\% \checkmark \text{ (a)}$$

## 22 Algebra बीजगणित

### AL-1 IDENTITY: VALUE DAALO

**PEHCHAN**  $x + 1/x = \dots \cdot a + b = \dots$ ,  $ab = \dots \cdot a + b + c = 0 \cdot x^3 + y^3 + z^3 - 3xyz$  · simplify / factorise

$$x + \frac{1}{x} = k \Rightarrow x^2 + \frac{1}{x^2} = k^2 - 2, \quad x^3 + \frac{1}{x^3} = k^3 - 3k$$

Shortcut: shart ke hisaab se aasaan number daalo ( $x = 1$ ,  $a = b = c$ ), phir options milao

| Sawaal mein ye ho             | Formula / kya karo                                                   | Example: value rakho → answer                                       |
|-------------------------------|----------------------------------------------------------------------|---------------------------------------------------------------------|
| ① $x + 1/x = k$               | $x^2 + 1/x^2 = k^2 - 2 \cdot x^3 + 1/x^3 = k^3 - 3k$                 | $k = 3 \rightarrow 7, 18$                                           |
| ② $x - 1/x = k$               | $x^2 + 1/x^2 = k^2 + 2 \cdot x^3 - 1/x^3 = k^3 + 3k$                 | $k = 2 \rightarrow 6, 14$                                           |
| ③ $x^2 - kx + 1 = 0$ (ya - 1) | x se bhaag: $x + 1/x = k$ (ya $x - 1/x = k$ )                        | $x^2 - 5x + 1 = 0 \rightarrow x + 1/x = 5$                          |
| ④ a + b aur ab diye           | $a^2 + b^2 = (a + b)^2 - 2ab \cdot a - b = \sqrt{((a + b)^2 - 4ab)}$ | $a + b = 7$ , $ab = 12 \rightarrow 25, 1$                           |
| ⑤ $a + b + c = 0$             | $a^3 + b^3 + c^3 = 3abc$                                             | $1, 2, -3 \rightarrow -18$                                          |
| ⑥ $a^3 + b^3 + c^3 - 3abc$    | $\frac{1}{2}(a + b + c)[(a - b)^2 + (b - c)^2 + (c - a)^2]$          | $555, 556, 557 \rightarrow \frac{1}{2} \times 1668 \times 6 = 5004$ |
| ⑦ Simplify / factorise / HCF  | $x = 1$ ya 2 daal ke sawaal aur options milao                        | $(x + 1)(x - 1)$ at $x = 2 \rightarrow 3 = x^2 - 1$<br>✓            |

BOOK SE (asli sawaal)

⑦ **SSC · Algebra Q338**  $(3x - 4)^3$  mein x ka coefficient?

$(a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3 \Rightarrow x$  wala term  $3 \times 3x \times 16 = 144x$  ✓ (c)

① **RRB English · Ch 22 Q64**  $x + 1/x = 3 \Rightarrow x^2 + 1/x^2$

$9 - 2 = 7$  ✓ (a)

### AL-2 EQUATION (LINEAR + QUADRATIC)

**PEHCHAN** solve, find x and y · roots / मूल · quadratic · remainder when divided by  $(x - a)$  · factor

$$ax^2 + bx + c = 0 : \quad \alpha + \beta = -\frac{b}{a}, \quad \alpha\beta = \frac{c}{a}, \quad D = b^2 - 4ac$$

$D > 0$  do real ·  $D = 0$  barabar ·  $D < 0$  real nahi

| Sawaal mein ye ho        | Formula / kya karo               | Example: value rakho → answer                               |
|--------------------------|----------------------------------|-------------------------------------------------------------|
| ① Do equation, do anjaan | Jodo / ghatao (ya options daalo) | $x + y = 10$ , $x - y = 2 \rightarrow 6, 4$                 |
| ② Roots / D              | $Jod = -b/a$ , $guna = c/a$      | $x^2 - 5x + 6 \rightarrow$ jod 5, $guna 6 \rightarrow 2, 3$ |
| ③ $(x - a)$ se shesh     | $x = a$ daalo                    | $(x^3 + 2) \div (x - 1) \rightarrow$ shesh $1 + 2 = 3$      |

BOOK SE (asli sawaal)

① **SSC · Algebra Q8**  $a + b + c = 7/12$ ,  $3a - 4b + 5c = 3/4$ ,  $7a - 11b - 13c = -7/12 \Rightarrow a + c?$

teen equation  $\Rightarrow a, b, c$  (Cramer)  $\Rightarrow a + c = 5/12$  ✓ (b)

① **RRB English · Ch 22 Q51**  $2x + 3y + 11 = 0$ ,  $6x + ky + 33 = 0$  coincident

$\frac{2}{6} = \frac{3}{k} \Rightarrow 9$  ✓ (b)

**AL-3 AP / GP (SERIES)****PEHCHAN** nth term · sum of n terms · progression / श्रेणी

$$\text{AP: } T_n = a + (n - 1)d, \quad S_n = \frac{n}{2}(a + l) \quad \text{GP: } T_n = a r^{n-1}$$

a = pehla · d = farak · r = guna · l = aakhri

| Sawaal mein ye ho | Formula / kya karo               | Example: value rakho → answer   |
|-------------------|----------------------------------|---------------------------------|
| ① AP term / jod   | $a + (n - 1)d \cdot n/2 (a + l)$ | 2, 5, 8 ... 10th = 29 · jod 155 |
| ② GP term         | $a \cdot r^{n-1}$                | 3, 6, 12 ... 5th = 48           |

**BOOK SE** (asli sawaal)① **SSC · Number System Q230**  $1\frac{1}{2} + 4\frac{1}{6} + 7\frac{1}{12} + 10\frac{1}{20} + \dots$  20 term?AP 1, 4, 7 ... 20 term =  $590 \cdot 1/(1 \cdot 2) + 1/(2 \cdot 3) + \dots = 20/21 \Rightarrow 12410/21$  ✓ (a)① **RRB English · Ch 22 Q1** pehle 25 odd ka jod $25^2 = 625$  ✓ (b)**AL-4 VENN (SETS)****PEHCHAN** both / दोनों · neither / कोई नहीं · only / केवल

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

Koi nahi = kul - n(A ∪ B)

| Sawaal mein ye ho | Formula / kya karo | Example: value rakho → answer                      |
|-------------------|--------------------|----------------------------------------------------|
| ① Dono / koi nahi | Upar wala formula  | 100 mein 60 chai, 50 coffee, 20 dono → 10 koi nahi |

**BOOK SE** (asli sawaal)① **RRB English · Ch 22 Q283** 14 ne A, 13 ne B, 2 ne dono $14 + 13 - 2 = 25$  ✓ (d)

## 23 Trigonometry त्रिकोणमिति

### TG-1 $\theta$ DAALO + VALUE TABLE

**PEHCHAN**  $\sin, \cos, \tan \dots$  ka maan · simplify / prove / equals ·  $\sec \theta + \tan \theta = k$  if  $\tan \theta = p/q$  · maximum / minimum value

| $\theta$                                                                                                                          | $0^\circ$ | $30^\circ$           | $45^\circ$           | $60^\circ$           | $90^\circ$ | $\sin^2 \theta + \cos^2 \theta = 1$ |
|-----------------------------------------------------------------------------------------------------------------------------------|-----------|----------------------|----------------------|----------------------|------------|-------------------------------------|
| sin                                                                                                                               | 0         | $\frac{1}{2}$        | $\frac{1}{\sqrt{2}}$ | $\frac{\sqrt{3}}{2}$ | 1          |                                     |
| cos                                                                                                                               | 1         | $\frac{\sqrt{3}}{2}$ | $\frac{1}{\sqrt{2}}$ | $\frac{1}{2}$        | 0          |                                     |
| tan                                                                                                                               | 0         | $\frac{1}{\sqrt{3}}$ | 1                    | $\sqrt{3}$           | $\infty$   |                                     |
| $\sec^2 - \tan^2 = 1 \cdot \operatorname{cosec}^2 - \cot^2 = 1 \cdot \sec = 1/\cos, \operatorname{cosec} = 1/\sin, \cot = 1/\tan$ |           |                      |                      |                      |            |                                     |

| Sawaal mein ye ho                                | Formula / kya karo                                            | Example: value rakho → answer                                                                |
|--------------------------------------------------|---------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| ① Simplify / equals                              | $\theta = 45^\circ$ (ya $30^\circ$ ) daalo, options milao     | $\sin^4 \theta - \cos^4 \theta$ at $45^\circ = 0 = \sin^2 \theta - \cos^2 \theta \checkmark$ |
| ② $\sec \theta + \tan \theta = k$                | $\sec \theta - \tan \theta = 1/k$                             | $k = 2 \rightarrow \sec \theta = (2 + 1/2)/2 = 5/4$                                          |
| ③ Ek ratio diya                                  | Triangle banao: p, b, h = $\sqrt{p^2 + b^2}$                  | $\tan \theta = 3/4 \rightarrow \sin 3/5, \cos 4/5$                                           |
| ④ $a \sin \theta + b \cos \theta$ max / min      | $\pm \sqrt{a^2 + b^2} \cdot a \sin^2 + b \cos^2$ : a se b tak | $3 \sin \theta + 4 \cos \theta \rightarrow \max 5$                                           |
| ⑤ $a \tan^2 \theta + b \cot^2 \theta$ ka minimum | $2\sqrt{ab}$                                                  | $9 \tan^2 + 4 \cot^2 \rightarrow 12$                                                         |
| ⑥ $\theta$ hatao / equation                      | Square karke jodo ( $\sin^2 + \cos^2 = 1$ )                   | $x = 3 \cos \theta, y = 3 \sin \theta \rightarrow x^2 + y^2 = 9$                             |

BOOK SE (asli sawaal)

① **SSC · Trigonometry Q57**  $\tan \theta + \cot \theta = 6 \Rightarrow \tan^2 \theta + \cot^2 \theta = ?$

$\tan + \cot = k \Rightarrow \tan^2 + \cot^2 = k^2 - 2 = 36 - 2 = 34 \checkmark$  (d)

① **RRB English · Ch 23 Q14**  $1/(1 + \sin \theta) + 1/(1 - \sin \theta)$

$\frac{2}{\cos^2 \theta} = 2 \sec^2 \theta \checkmark$  (c)

### TG-2 ANGLE BADLO ( $90^\circ$ JODI)

**PEHCHAN**  $\sin 37^\circ / \cos 53^\circ \cdot \tan 1^\circ \tan 2^\circ \dots \tan 89^\circ \cdot \sin A = \cos B \cdot \sin(A + B), \cos 2\theta \cdot \text{radian}$

$$A + B = 90^\circ \Rightarrow \sin A = \cos B, \quad \tan A \cdot \tan B = 1$$

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B \cdot \pi \text{ rad} = 180^\circ$$

| Sawaal mein ye ho                                             | Formula / kya karo                                                  | Example: value rakho → answer                                             |
|---------------------------------------------------------------|---------------------------------------------------------------------|---------------------------------------------------------------------------|
| ① $90^\circ$ wali jodi                                        | $\sin A = \cos B \Rightarrow A + B = 90^\circ$                      | $\tan x = \cot(65^\circ + 9x) \rightarrow 10x = 25 \rightarrow 2.5^\circ$ |
| ② $\sin^2 1^\circ + \sin^2 2^\circ + \dots + \sin^2 89^\circ$ | Jodi ka jod 1 $\rightarrow \text{kul} = 89/2$                       | $= 44.5$                                                                  |
| ③ $A + B / 2\theta$                                           | $\sin(A + B), \tan(A + B)$ formula                                  | $\tan 15^\circ = 2 - \sqrt{3}$                                            |
| ④ Degree $\leftrightarrow$ radian                             | Degree $\times \pi/180 \cdot \text{arc} = r \times \theta$ (radian) | $60^\circ = \pi/3$                                                        |

BOOK SE (asli sawaal)

SSC • Trigonometry Q269 cosec  $39^\circ = x \Rightarrow 1/\text{cosec}^2 51^\circ + \sin^2 39^\circ + \tan^2 51^\circ - 1/(\sin^2 51^\circ \cdot \sec^2 39^\circ)$  $51^\circ = 90^\circ - 39^\circ : \sin^2 51^\circ + \sin^2 39^\circ = 1, \tan^2 51^\circ = \cot^2 39^\circ = x^2 - 1, \text{aakhri} = 1 \Rightarrow x^2 - 1 \checkmark \text{ (c)}$ ① RRB English • Ch 23 Q155  $\tan 1^\circ \cdot \tan 2^\circ \cdots \tan 89^\circ$ jodiyan = 1  $\checkmark \text{ (c)}$ **TG-3 UNCHAI – DOORI****PEHCHAN** angle of elevation / उन्नयन कोण • angle of depression / अवनमन • tower, shadow, ladder, kite

$$\tan \theta = \frac{\text{Unchai}}{\text{Doori}} \quad 30^\circ : \frac{d}{\sqrt{3}} \quad 45^\circ : d \quad 60^\circ : d\sqrt{3}$$

Seedhi / dori di ho: unchai = lambai  $\times \sin \theta$ 

| Sawaal mein ye ho                         | Formula / kya karo                                                                                      | Example: value rakho → answer             |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------|-------------------------------------------|
| ① Tower / chhaya                          | $h = d \times \tan \theta$                                                                              | $d = 30, 60^\circ \rightarrow 30\sqrt{3}$ |
| ② Paas chala d (kon $\alpha$ se $\beta$ ) | $h = d \div (\cot \alpha - \cot \beta) \cdot 30^\circ \rightarrow 60^\circ : h = (\sqrt{3}/2) \times d$ | $d = 20 \rightarrow 10\sqrt{3}$           |
| ③ Dori / seedhi                           | $h = l \times \sin \theta$                                                                              | dori 100 m, $30^\circ \rightarrow 50$ m   |

BOOK SE (asli sawaal)

① SSC • Height and Distance Q10 Patang 90 m unchi, dori  $30^\circ$  par  $\Rightarrow$  dori ki lambai?dori l:  $h = l \sin \theta \Rightarrow l = 90 \div \sin 30^\circ = 180 \checkmark \text{ (a)}$ 

① RRB English • Ch 23 Q174 18 m aur 32 m se complementary angles

 $\sqrt{18 \cdot 32} = 24 \checkmark \text{ (d)}$

## 24 Co-ordinate Geometry निर्देशांक ज्यामिति

### CG-1 POINTS: DOORI • BEECH • AREA

**PEHCHAN** distance between points / दूरी • midpoint / ratio • area of triangle (x, y) • collinear

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \quad \text{Area} = \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

$$\text{Midpoint} = \text{dono ka average} \cdot m : n \text{ mein} = (mx_2 + nx_1)/(m+n)$$

| Sawaal mein ye ho               | Formula / kya karo                                                                                            | Example: value rakho → answer        |
|---------------------------------|---------------------------------------------------------------------------------------------------------------|--------------------------------------|
| ① Doori                         | Upar wala formula                                                                                             | (1, 2), (4, 6) → 5                   |
| ② Midpoint / ratio / reflection | $(mx_2 + nx_1) \div (m+n)$ • reflection: x-axis (x, -y), y-axis (-x, y), line $y = k \rightarrow (x, 2k - y)$ | (2, 4), (8, 10) ka midpoint → (5, 7) |
| ③ Triangle ka area              | Formula (0 aaye to ek line par)                                                                               | (0,0), (4,0), (0,3) → 6              |

BOOK SE (asli sawaal)

② **SSC • Co-ordinate Geometry Q2** (5, -3) ka line  $y = 3$  mein reflection?

line  $y = k$  mein reflection:  $(x, 2k - y) = (5, 6 + 3) = (5, 9)$  ✓ (d)

① **RRB English • Ch 24 Q39** square ke consecutive vertices (2, -2), (5, 2)

$\sqrt{9 + 16} = 5$  ✓ (d)

### CG-2 LINE + CIRCLE

**PEHCHAN** slope / ढल • equation of line • parallel / perpendicular • intercept, area with axes • circle centre, radius

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad \parallel: m_1 = m_2 \quad \perp: m_1 m_2 = -1$$

$$ax + by = c: \text{x-intercept } c/a, \text{ y-intercept } c/b$$

| Sawaal mein ye ho           | Formula / kya karo                                                                                          | Example: value rakho → answer                                |
|-----------------------------|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| ① Slope / line              | $ax + by + c = 0 \rightarrow m = -a/b$ • point aur slope: $y - y_1 = m(x - x_1)$ • y-axis par $x = 0$ rakho | $3x + 4y = 12 \rightarrow m = -3/4$                          |
| ② Line aur axes ka triangle | Area = $\frac{1}{2} \times (c/a) \times (c/b)$                                                              | $3x + 4y = 12 \rightarrow \frac{1}{2} \times 4 \times 3 = 6$ |
| ③ Circle                    | $(x - h)^2 + (y - k)^2 = r^2$                                                                               | centre (1, -2), r 4 $\rightarrow x^2 + y^2 - 2x + 4y = 11$   |

BOOK SE (asli sawaal)

① **SSC • Co-ordinate Geometry Q10** (3, -5) se guzarti, slope 2 wali line?

$y - y_1 = m(x - x_1): y + 5 = 2(x - 3) \Rightarrow 2x - y - 11 = 0$  ✓ (c)

① **RRB English • Ch 24 Q30**  $3x - 2y - 11 = 0$  aur  $x + y = 7 \Rightarrow 3\alpha + 5\beta$

(5, 2)  $\Rightarrow 25$  ✓ (c)

## 25 Geometry ज्यामिति

### GE-1 KON (ANGLES)

**PEHCHAN** angle / कोण · parallel lines · triangle angles, exterior angle · bisectors meet at O · polygon: interior / exterior · parallelogram, cyclic quadrilateral

Seedhi rekha =  $180^\circ$       Triangle =  $180^\circ$       Bahari kon = door ke do andar ke kon ka jod

| Sawaal mein ye ho                              | Formula / kya karo                                                                                      | Example: value rakho → answer                                                |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| ① <b>Lines &amp; angles</b>                    | Z (alternate) barabar · F (corresponding) barabar · co-interior jod $180^\circ$                         | ek kon $70^\circ \rightarrow$ alternate $70^\circ$ , co-interior $110^\circ$ |
| ② <b>Bisectors milte hain</b> ( $\angle BOC$ ) | Incentre $90^\circ + A/2$ · excentre $90^\circ - A/2$ · circumcentre $2A$ · orthocentre $180^\circ - A$ | $A = 60^\circ \rightarrow 120^\circ, 60^\circ, 120^\circ, 120^\circ$         |
| ③ <b>Altitude aur bisector ke beech</b>        | $\frac{1}{2}(B - C)$ · B ka andar + C ka bahar bisector $\rightarrow A/2$                               | $B 72^\circ, C 26^\circ \rightarrow 23^\circ$                                |
| ④ <b>Polygon</b>                               | Exterior = $360^\circ/n$ · interior = $180^\circ - 360^\circ/n$ · diagonals $n(n-3)/2$                  | 8 bhuja $\rightarrow 45^\circ, 135^\circ, 20$                                |
| ⑤ <b>Chaar bhuja / cyclic</b>                  | Jod $360^\circ$ · cyclic: saamne ke kon ka jod $180^\circ$ · rhombus: $side^2 = (d_1/2)^2 + (d_2/2)^2$  | cyclic, $\angle A = 70^\circ \rightarrow \angle C = 110^\circ$               |

BOOK SE (asli sawaal)

③ **SSC · Geometry Q144** Bahar ke kon ke bisectors O par,  $\angle BOC = 57^\circ \Rightarrow \angle A?$

bahar ke bisectors:  $\angle BOC = 90^\circ - A/2 = 57^\circ \Rightarrow A = 66^\circ$  ✓ (a)

① **RRB English · Ch 25 Q236** supplementary, bada  $36^\circ$  zyada

$\frac{180-36}{2} = 72^\circ$  ✓ (a)

### GE-2 TRIANGLE KI LAMBAI

**PEHCHAN** Pythagoras · similar triangles / समरूप · angle bisector · median, centroid / माधिका · perpendicular on hypotenuse · inradius, circumradius

$$c^2 = a^2 + b^2 \quad \text{Similar: sides } k \Rightarrow \text{area } k^2$$

$$30-60-90 \rightarrow 1 : \sqrt{3} : 2 \quad 45-45-90 \rightarrow 1 : 1 : \sqrt{2}$$

| Sawaal mein ye ho                                         | Formula / kya karo                                                                           | Example: value rakho → answer             |
|-----------------------------------------------------------|----------------------------------------------------------------------------------------------|-------------------------------------------|
| ① <b>Similar / BPT / bisector</b>                         | Area ratio = side <sup>2</sup> · bisector: $BD/DC = AB/AC$                                   | sides $3 : 2 \rightarrow$ area $9 : 4$    |
| ② <b>Median / centroid</b>                                | Centroid $2 : 1$ · medians $\rightarrow$ 6 barabar area · $m^2 = (2b^2 + 2c^2 - a^2)/4$      | $b = c = 5, a = 6 \rightarrow$ median $4$ |
| ③ <b>Right triangle: hyp par perpendicular</b>            | $AD = ab/c$ · $AD^2 = BD \times DC$                                                          | $6, 8, 10 \rightarrow AD = 4.8$           |
| ④ <b><math>120^\circ / 135^\circ</math> wala triangle</b> | $a^2 = b^2 + c^2 - 2bc \cos A$ ( $120^\circ \rightarrow +bc$ )                               | $b 3, c 5, A 120^\circ \rightarrow a = 7$ |
| ⑤ <b>Inradius r, circumradius R</b>                       | Right: $r = (a + b - c)/2$ , $R = c/2$ · equilateral: $R = a/\sqrt{3}$ , $r = a/(2\sqrt{3})$ | $6, 8, 10 \rightarrow r 2, R 5$           |

BOOK SE (asli sawaal)

SSC • Geometry Q49 DE || BC, AD = 2, BD = 5.2, AC = 9 ⇒ AE?

BPT: AD/AB = AE/AC ⇒ 2/7.2 = x/9 ⇒ x = 2.5 ✓ (d)

① RRB English • Ch 25 Q2 PQ 12, PR 15, QR 18; bisector S ⇒ SR

 $18 \cdot \frac{15}{27} = 10$  ✓ (c)**GE-3 CIRCLE: CHORD • TANGENT****PEHCHAN** chord / जीवा • tangent / स्पर्श रेखा • angle at centre • two circles, common tangent / chord • secant meets outside

$$r^2 = d^2 + \left(\frac{\text{chord}}{2}\right)^2 \quad \text{Tangent}^2 = d^2 - r^2 \quad \angle \text{centre} = 2 \times \angle \text{paridhi}$$

d = centre se doori

| Sawaal mein ye ho              | Formula / kya karo                                                                                   | Example: value rakho → answer |
|--------------------------------|------------------------------------------------------------------------------------------------------|-------------------------------|
| ① Chord / kon                  | $r^2 = d^2 + (\text{chord}/2)^2$ • semicircle mein kon $90^\circ$                                    | r 13, chord 24 → d = 5        |
| ② Tangent                      | Bahar se do tangent barabar • tangents ka kon + centre ka kon = $180^\circ$ • $PT^2 = PA \times PB$  | d 13, r 5 → tangent 12        |
| ③ Do circles ki common tangent | Direct $\sqrt{(d^2 - (r_1 - r_2)^2)}$ • transverse $\sqrt{(d^2 - (r_1 + r_2)^2)}$                    | d 10, r 4, 2 → transverse 8   |
| ④ Common chord / secant        | $PA \times PB = d^2 - r^2$ • common chord = $2\sqrt{(r_1^2 - x^2)}$ , $x = (d^2 + r_1^2 - r_2^2)/2d$ | r 9, 12, d 15 → 14.4          |

BOOK SE (asli sawaal)

② SSC • Geometry Q559 Do radius ke beech  $130^\circ$  ⇒ dono tangents ke beech kon?tangents ka kon + centre ka kon =  $180^\circ$  ⇒  $180 - 130 = 50^\circ$  ✓ (c)

① RRB English • Ch 25 Q129 chord 24, centre se 5

 $\sqrt{144 + 25} = 13$  ✓ (a)

## 26 Statistics & Probability सांख्यिकी और प्रायिकता

### SP-1 MEAN – MEDIAN – MODE

**PEHCHAN** mean / माध्य · median / माधिका · mode / बहुलक · frequency table

$$\text{Mean} = \frac{\sum fx}{\sum f} \quad \text{Mode} = 3 \text{ Median} - 2 \text{ Mean}$$

Median = sort karke beech wala

| Sawaal mein ye ho | Formula / kya karo                            | Example: value rakho → answer |
|-------------------|-----------------------------------------------|-------------------------------|
| ① Mean            | $\Sigma fx \div \Sigma f$                     | 2, 4, 6, 8 → 5                |
| ② Median          | Sort karo, beech wala (n even: do ka average) | 3, 1, 7, 5 → 1, 3, 5, 7 → 4   |
| ③ Mode            | Sabse zyada baar                              | 2, 3, 3, 5 → 3                |
| ④ Teeno ka rishta | Mode = 3 Median – 2 Mean                      | median 35, mode 27 → mean 39  |

BOOK SE (asli sawaal)

② **SSC · Number System Q275** Wickets 0, 3, 2, 1, 5, 3, 4, 5, 5, 2, 2, 0, 0, 1, 2 ka median?

sort: 0,0,0,1,1,2,2,2,2,3,3,4,5,5,5 ⇒ 8th = 2 (median) ✓ (d)

① **RRB English · Ch 26 Q1** 10 values ka mean, har ek +10

14 + 10 = 24 ✓ (d)

### SP-2 RANGE · SD

**PEHCHAN** range / परास · variance / प्रसरण · standard deviation / मानक विचलन

$$\text{Range} = \max - \min \quad \sigma^2 = \frac{\sum x^2}{n} - \bar{x}^2$$

CV =  $\sigma \div \text{mean} \times 100$

| Sawaal mein ye ho | Formula / kya karo                                                                  | Example: value rakho → answer                             |
|-------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------|
| ① Range / SD      | $\sigma = \sqrt{\text{variance}} \cdot \text{CV} = \sigma / \text{mean} \times 100$ | variance 324 → $\sigma$ 18 · mean 24, $\sigma$ 6 → CV 25% |

BOOK SE (asli sawaal)

① **RRB English · Ch 26 Q134** mean 25, SD 5 ⇒ CV

20% ✓ (c)

### SP-3 PROBABILITY

**PEHCHAN** probability / प्रायिकता · coin, dice, cards · balls drawn

$$P = \frac{\text{Chahiye wale tarike}}{\text{Kul tarike}}$$

2 dice = 36 · cards = 52 · P(nahi) = 1 – P

| Sawaal mein ye ho     | Formula / kya karo | Example: value rakho → answer |
|-----------------------|--------------------|-------------------------------|
| ① Coin / dice / cards | Favourable ÷ total | 2 dice, jod 7 → 6/36 = 1/6    |

**BOOK SE** (asli sawaal)① **RRB English · Ch 26 Q254** 3 coins, kam se kam ek head

$$1 - \frac{1}{8} = \frac{7}{8} \checkmark \text{ (d)}$$

## 27 Data Interpretation आंकड़ों का विश्लेषण

### DI-1 GRAPH / TABLE PADHO

**PEHCHAN** pie chart · bar graph · table / तालिका · line graph · percentage increase, ratio, average

$$\% \text{ badlav} = \frac{\text{Naya} - \text{Purana}}{\text{Purana}} \times 100 \quad 1\% = 3.6^\circ$$

Sirf zaroori numbers uthao

| Sawaal mein ye ho              | Formula / kya karo                                | Example: value rakho → answer             |
|--------------------------------|---------------------------------------------------|-------------------------------------------|
| ① <b>Pie chart</b>             | Hissa = $\text{kon} \div 360 \times \text{total}$ | $90^\circ$ of 2000 → 500                  |
| ② <b>% badlav / kitna %</b>    | Upar wala formula (base = purana)                 | 110 → 150 → 36.36%                        |
| ③ <b>Ratio / farak / total</b> | Zaroori numbers jodo, phir HCF se chhota          | $(20 + 30) : (40 + 60) \rightarrow 1 : 2$ |
| ④ <b>Average / max-min</b>     | Jod ÷ ginti · max / min seedha padho              | 12, 15, 18 → avg 15                       |

**BOOK SE** (asli sawaal)

③ **SSC · Data Interpretation Q122** (A 2014 + C 2013) : (B 2016 + E 2015) = (48 + 36) : (56 + 35)?  
 $(48 + 36) : (56 + 35) = 84 : 91 = 12 : 13$  ✓ (d)

① **RRB English · Ch 27 Q3** Delhi 30% ⇒ central angle  
 $30 \times 3.6 = 108^\circ$  ✓ (c)

## 28 Miscellaneous विविध

### MS-1 BACHA HISSA + EK KA KITNA

**PEHCHAN** of the remaining / बाकी का · map scale 1 : n · unitary / ऐकिक · puzzle: doubling, digits

$$\text{Poora} = \frac{\text{Bacha hua}}{(1-a)(1-b)\dots} \quad 1 \text{ ka} = \frac{\text{Kul}}{\text{Ginti}}$$

Puzzle: aakhir se ulta chalo ya options daalo

|   | Sawaal mein ye ho      | Formula / kya karo                | Example: value rakho → answer                                         |
|---|------------------------|-----------------------------------|-----------------------------------------------------------------------|
| ① | Hisse diye, bacha diya | Bacha ÷ (bache fractions ka guna) | $\frac{1}{2}$ kharch, phir baaki ka $\frac{1}{3}$ , bacha ₹200 → ₹600 |
| ② | Scale / ek ka kitna    | Pehle 1 ka, phir guna             | 5 pen ₹40 → 8 pen ₹64                                                 |
| ③ | Doubling / pattern     | Poora t din → aadha (t - 1) din   | lily 30 din mein poora → aadha 29                                     |
| ④ | Digit rule             | Har digit badlo, phir jo poocha   | 835647: odd + 2, even - 3 → 557319                                    |

BOOK SE (asli sawaal)

① **RRB English · Ch 28 Q7**  $\frac{1}{4}$  bhara, 10 cup daale ⇒  $\frac{3}{4}$   
 $\frac{\frac{10}{\frac{3}{4}-\frac{1}{4}}}{\frac{3}{4}-\frac{1}{4}} = 20$  ✓ (b)

# Revision: Saare 53 Formula Ek Jagah

## 1. Number System

(NS-1) BHAAG – SHESH

$$N = D \times Q + R$$

(NS-2) GINTI + FACTORS

$$N = p^a q^b \Rightarrow \text{Factors} = (a+1)(b+1)$$

(NS-3) DO NUMBER + SERIES

$$\text{Bada} = \frac{S+D}{2} \quad \text{Chhota} = \frac{S-D}{2}$$

## 2. Decimal Fractions

(DF-1) DECIMAL ↔ FRACTION

$$0.\overline{ab} = \frac{ab}{99} \quad 0.a\overline{b} = \frac{ab-a}{90}$$

(DF-2) FRACTION: BADA-CHHOTA + JOD

$$\frac{a}{b} \text{ vs } \frac{c}{d} : \quad a \times d \text{ vs } b \times c$$

## 3. Indices and Surds

(IS-1) POWER (GHAAT)

$$a^m \cdot a^n = a^{m+n} \quad \frac{a^m}{a^n} = a^{m-n} \quad (a^m)^n = a^{mn} \quad a^{-n} = \frac{1}{a^n}$$

(IS-2) √ ROOT + SURD

$$\sqrt{a^2 b} = a\sqrt{b} \quad (a + \sqrt{b})(a - \sqrt{b}) = a^2 - b$$

## 4. Simplification

(SIM-1) BODMAS + “?”

$$\text{Bracket} \rightarrow \text{of} \rightarrow \div \rightarrow \times \rightarrow + \rightarrow -$$

(SIM-2) IDENTITY SHORTCUT

$$a^2 - b^2 = (a+b)(a-b) \quad (a+b)^2 - (a-b)^2 = 4ab$$

## 5. LCM & HCF

(LH-1) HCF – LCM

$$\text{HCF} \times \text{LCM} = \text{Pehla} \times \text{Doosra}$$

(LH-2) SAATH KAB / SABSE BADA

$$\text{Saath / sabse chhota} \Rightarrow \text{LCM} \quad \text{Sabse bada / naap} \Rightarrow \text{HCF}$$

## 6. Percentage

(PC-1) % = HISSA

$$x\% \text{ of } Y = \frac{x \times Y}{100}$$

(PC-2) BADHA – GHATA (MULTIPLIER)

$$\text{Naya} = \text{Purana} \times \frac{100 \pm x}{100}$$

## 7. Profit & Loss

(PL-1) CP → SP (PROFIT / LOSS)

$$SP = CP \times \frac{100 \pm P}{100}$$

(PL-2) GINTI · RATE · NAKLI BAAT

$$P\% = \frac{y-x}{x} \times 100$$

**8. Discount**

(DS-1) DISCOUNT

$$SP = MP \times \frac{100 - d}{100}$$

**9. Ratio & Proportion**

(RP-1) "1 HISSA" (RATIO)

$$1 \text{ hissa} = \frac{\text{Di hui value}}{\text{uske parts}}$$

(RP-2) CROSS GUNA (PROPORTION)

$$\frac{a}{b} = \frac{c}{d} \iff a \times d = b \times c$$

**10. Partnership**

(PT-1) PROFIT = PAISA × SAMAY

$$A : B = \text{Paissa}_A \times \text{Mahine}_A : \text{Paissa}_B \times \text{Mahine}_B$$

**11. Work & Time**

(WT-1) KAAM = LCM

$$\text{Kul kaam} = \text{LCM}(\text{din}) \quad \text{Saath ke din} = \frac{a \times b}{a + b}$$

(WT-2) LOG × DIN × GHANTE

$$\frac{M_1 D_1 H_1}{W_1} = \frac{M_2 D_2 H_2}{W_2}$$

**12. Alligation / Mixture**

(MX-1) ALLIGATION (CROSS)

$$\frac{\text{Sasta}}{\text{Mehenga}} = \frac{d - m}{m - c}$$

(MX-2) PAANI MILAO / NIKAALO

$$C_1 V_1 = C_2 V_2 \quad \text{Bacha} = \text{Shuru} \times \left(1 - \frac{x}{V}\right)^n$$

**13. Pipe & Cistern**

(PP-1) NAL – TANKI

$$\frac{1}{T} = \frac{1}{a} + \frac{1}{b} - \frac{1}{c}$$

**14. Simple Interest**

(SI-1) SIMPLE INTEREST (P·R·T)

$$SI = \frac{P \times R \times T}{100} \quad A = P + SI$$

**15. Compound Interest**

(CI-1) COMPOUND INTEREST

$$A = P \left(1 + \frac{R}{100}\right)^n$$

(CI-2) CI – SI FARAK

$$\mathbf{2 \text{ saal: } P \left(\frac{R}{100}\right)^2} \quad \mathbf{3 \text{ saal: } P \left(\frac{R}{100}\right)^2 \left(3 + \frac{R}{100}\right)}$$

**16. Problems on Age**

(AG-1) UMAR (AGE)

$$\text{Umar} = ax, bx \quad \frac{ax + t}{bx + t} = \frac{c}{d} \Rightarrow x = \frac{t(c - d)}{ad - bc}$$

**17. Average**

$$\text{AV-1} \quad \text{AVERAGE} = \text{TOTAL} \div n$$

$$\text{Total} = \text{Average} \times n$$

**18. Speed, Time & Distance**

$$\text{ST-1} \quad \text{DOORI} = \text{SPEED} \times \text{SAMAY}$$

$$D = S \times T \quad \text{km/h} \times \frac{5}{18} = \text{m/s}$$

$$\text{ST-2} \quad \text{AVERAGE} + \text{RELATIVE SPEED}$$

$$\text{Avg} = \frac{2xy}{x+y} \quad \text{Relative} = S_1 + S_2 \text{ (aamne)}, \quad S_1 - S_2 \text{ (peeche)}$$

**19. Train**

$$\text{TR-1} \quad \text{TRAIN}$$

$$\text{Samay} = \frac{L_{\text{train}} + L_{\text{cheez}}}{\text{Relative speed}}$$

**20. Boat & Stream**

$$\text{BS-1} \quad \text{NAAV} - \text{DHAARA}$$

$$B = \frac{D+U}{2} \quad S = \frac{D-U}{2}$$

**21. Mensuration**

$$\text{MN-1} \quad \text{2-D AREA TABLE}$$

| Shape       | Area                              | Perimeter   |
|-------------|-----------------------------------|-------------|
| Triangle    | $\frac{1}{2} b h$                 | $a + b + c$ |
| Equilateral | $\frac{\sqrt{3}}{4} a^2$          | $3a$        |
| Rectangle   | $l \times b$                      | $2(l + b)$  |
| Square      | $a^2$                             | $4a$        |
| Rhombus     | $\frac{1}{2} d_1 d_2$             | $4a$        |
| Trapezium   | $\frac{1}{2} (a + b) h$           |             |
| Circle      | $\pi r^2$                         | $2\pi r$    |
| Hexagon     | $6 \times \frac{\sqrt{3}}{4} a^2$ | $6a$        |

$$\text{MN-2} \quad \text{3-D VOLUME TABLE}$$

| Shape      | Volume                             | Surface (TSA)     |
|------------|------------------------------------|-------------------|
| Cube       | $a^3$                              | $6a^2$            |
| Cuboid     | $l b h$                            | $2(lb + bh + hl)$ |
| Cylinder   | $\pi r^2 h$                        | $2\pi r(r + h)$   |
| Cone       | $\frac{1}{3} \pi r^2 h$            | $\pi r(r + l)$    |
| Sphere     | $\frac{4}{3} \pi r^3$              | $4\pi r^2$        |
| Hemisphere | $\frac{2}{3} \pi r^3$              | $3\pi r^2$        |
| Prism      | $\text{Base} \times h$             |                   |
| Pyramid    | $\frac{1}{3} \text{Base} \times h$ |                   |

$$\text{MN-3} \quad \text{SIZE BADLA}$$

$$\text{Lambai} \times k \Rightarrow \text{Area} \times k^2, \quad \text{Volume} \times k^3$$

**22. Algebra****(AL-1)** IDENTITY: VALUE DAALO

$$x + \frac{1}{x} = k \Rightarrow x^2 + \frac{1}{x^2} = k^2 - 2, \quad x^3 + \frac{1}{x^3} = k^3 - 3k$$

**(AL-2)** EQUATION (LINEAR + QUADRATIC)

$$ax^2 + bx + c = 0 : \quad \alpha + \beta = -\frac{b}{a}, \quad \alpha\beta = \frac{c}{a}, \quad D = b^2 - 4ac$$

**(AL-3)** AP / GP (SERIES)

$$\mathbf{AP:} T_n = a + (n-1)d, \quad S_n = \frac{n}{2}(a+l) \quad \mathbf{GP:} T_n = ar^{n-1}$$

**(AL-4)** VENN (SETS)

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

**23. Trigonometry****(TG-1)** θ DAALO + VALUE TABLE

| $\theta$ | $0^\circ$ | $30^\circ$           | $45^\circ$           | $60^\circ$           | $90^\circ$ |
|----------|-----------|----------------------|----------------------|----------------------|------------|
| sin      | 0         | $\frac{1}{2}$        | $\frac{1}{\sqrt{2}}$ | $\frac{\sqrt{3}}{2}$ | 1          |
| cos      | 1         | $\frac{\sqrt{3}}{2}$ | $\frac{1}{\sqrt{2}}$ | $\frac{1}{2}$        | 0          |
| tan      | 0         | $\frac{1}{\sqrt{3}}$ | 1                    | $\sqrt{3}$           | $\infty$   |

$$\sin^2 \theta + \cos^2 \theta = 1$$
**(TG-2)** ANGLE BADLO ( $90^\circ$  JODI)

$$A + B = 90^\circ \Rightarrow \sin A = \cos B, \quad \tan A \cdot \tan B = 1$$

**(TG-3)** UNCHAI - DOORI

$$\tan \theta = \frac{\text{Unchai}}{\text{Doori}} \quad 30^\circ : \frac{d}{\sqrt{3}} \quad 45^\circ : d \quad 60^\circ : d\sqrt{3}$$

**24. Co-ordinate Geometry****(CG-1)** POINTS: DOORI · BEECH · AREA

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \quad \text{Area} = \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

**(CG-2)** LINE + CIRCLE

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad \parallel : m_1 = m_2 \quad \perp : m_1 m_2 = -1$$

**25. Geometry****(GE-1)** KON (ANGLES)

$$\text{Seedhi rekha} = 180^\circ \quad \text{Triangle} = 180^\circ \quad \text{Bahari kon} = \text{door ke do andar ke kon ka jod}$$

**(GE-2)** TRIANGLE KI LAMBAI

$$c^2 = a^2 + b^2 \quad \text{Similar: sides } k \Rightarrow \text{area } k^2$$

**(GE-3)** CIRCLE: CHORD · TANGENT

$$r^2 = d^2 + \left(\frac{\text{chord}}{2}\right)^2 \quad \text{Tangent}^2 = d^2 - r^2 \quad \angle \text{centre} = 2 \times \angle \text{paridhi}$$

**26. Statistics & Probability****(SP-1)** MEAN - MEDIAN - MODE

$$\text{Mean} = \frac{\sum fx}{\sum f} \quad \text{Mode} = 3 \text{ Median} - 2 \text{ Mean}$$

**(SP-2)** RANGE · SD

$$\text{Range} = \max - \min \quad \sigma^2 = \frac{\sum x^2}{n} - \bar{x}^2$$

**(SP-3)** PROBABILITY

$$P = \frac{\text{Chahiye wale tarike}}{\text{Kul tarike}}$$

**27. Data Interpretation****(DI-1)** GRAPH / TABLE PADHO

$$\% \text{ badlav} = \frac{\text{Naya} - \text{Purana}}{\text{Purana}} \times 100 \quad 1\% = 3.6^\circ$$

**28. Miscellaneous****(MS-1)** BACHA HISSA + EK KA KITNA

$$\text{Poora} = \frac{\text{Bacha hua}}{(1-a)(1-b)\dots} \quad 1 \text{ ka} = \frac{\text{Kul}}{\text{Ginti}}$$